Error Propagation in Cascaded Speech Translation under Acoustic Channel Degradations: A Black-Box Case Study of Commercial Speech Intelligence APIs

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Abstract

Cloud speech services are black boxes, yet their acoustic channels are rarely clean. CascadeBench is a seeded black-box benchmark measuring how fourteen degradations from seven acoustic families propagate through commercial speech APIs: five recognition systems in three product classes of one provider, three cascaded and two end-to-end translation paths, on a 480-sentence bilingual corpus with an eight-level babble sweep (20 to −10 dB). Recognition error rises as a knee-shaped function of noise: the audio-LLM systems tolerate 2.2–2.5 dB (Mandarin) and 4.4–4.8 dB (English) more noise than the specialised streaming system before reaching ten percent error, then collapse below 0 dB into insertion-heavy hallucination above 100 percent character error. Cascade loss follows an empirical pass-through of 0.4 to 0.5 BLEU points per percentage point of recognition error; a three-point re-fit on a second vendor's engine reproduces it at the same order of magnitude, and engine-independence of the coefficient is not established. A 150-utterance real-speech validation with a six-level babble ladder reproduces the knee and hallucination structure, finds measured-RIR rooms far milder than the synthetic stress case, and bounds the audio-LLM advantage to the near-knee regime. End-to-end translation is not automatically more robust: the omni-modal path is the most fragile end-to-end

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path and degrades more than the LLM-frontend cascades in point estimate, with a reference-dependent margin. Three-reference scoring with one family-disjoint reference shows relative-loss rankings are stable up to a sub-point inversion under the same-engine reference. The recipe, corpus, degraded audio, and 43,046 logged API responses are released.

Keywords: speech recognition, speech translation, robustness benchmark, acoustic channel degradations, cascaded systems, cloud APIs

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1. Introduction

Speech has become a primary interface to cloud intelligence. Voice assistants, meeting transcription products, live captioning services, and multilingual communication platforms are routinely assembled from commercial automatic speech recognition (ASR) and machine translation (MT) interfaces offered by large cloud providers. In a typical deployment, audio captured by a far-field microphone or transported over a communication channel is first converted to text by an ASR service, and the resulting text is then passed to a text translation engine whenever the application crosses a language boundary. This cascade architecture dominates industrial practice because it allows each stage to be selected, replaced, and billed independently, and because the best available components at each stage are often operated by different providers. An alternative architecture, end-to-end speech translation (E2E ST), maps source-language audio directly to target-language text and has matured rapidly with large foundation models.

The acoustic pathway between a speaker and these services is rarely pristine. Speech arrives mixed with competing talkers, shaped by room reverberation, band-limited by telephone channels, compressed by low-bitrate codecs, or corrupted by packet loss on unstable network links. The robustness of ASR systems to such degradations has been a central concern of the speech research community for decades, and controlled robustness corpora have driven substantial progress. What has changed is the identity of the systems that now dom-

inate deployment. Commercial speech services are increasingly built on large language model (LLM) based ASR, on general-purpose multimodal foundation models that accept audio alongside text, and on streaming speech translation models, none of which can be fine-tuned, inspected, or benchmarked offline by their consumers. A systems integrator who composes such services into a translation pipeline has no principled way to predict how recognition errors induced by an acoustic channel will propagate into the final translation, or whether a cascade or an end-to-end design is the safer choice under the conditions of a given deployment.

The evaluation gap is not academic. Deployment decisions are made weekly by integrators who must choose among service tiers and architectural patterns with nothing but clean-audio demo pages and marketing claims, while the operating conditions of real audio pathways, conference rooms, vehicles, telephone trunks, and congested wireless links, are precisely the conditions that clean demos do not exercise. A deployment that selects a recognition frontend on clean-audio accuracy alone can be quietly losing most of its translation quality in the field, and a deployment that assumes end-to-end robustness can be more fragile than the cascade it replaced. Closing this gap requires measurement that is controlled at the acoustic layer, complete at the pipeline layer, and reproducible at the service layer, and that is the design brief of the benchmark presented here.

This gap motivates the present study. Academic robustness benchmarks such as Aurora [1] and the CHiME challenge series [2] evaluate open recognizers on fixed noise corpora, but they do not cover the black-box commercial systems that now carry production traffic, they rarely extend beyond recognition to the downstream translation stage, and they cannot be re-run on new model versions without access to model internals. Conversely, studies that compare commercial ASR services report error rates on fixed test sets but typically consider clean audio and stop at the transcription boundary. The propagation of acoustic degradation through a complete speech translation pipeline, comparing product classes and both cascade and end-to-end designs under identical and

fully controlled channel conditions, has to our knowledge not been quantified for the current generation of speech APIs.

This paper presents CascadeBench, a black-box benchmark designed to answer four questions. First, how sensitive are the main product classes of commercial ASR, namely specialised streaming models, LLM-based recognisers, and general-purpose multimodal models, to a controlled grid of channel degradations spanning seven acoustic families? Second, how does recognition accuracy scale with noise severity across the full range from mild to destructive, and where does each class fail? Third, when recognition errors flow into a downstream translation engine, does the cascade amplify them, and how does the cascade compare with end-to-end speech translation under identical conditions? Fourth, what are the latency consequences of each choice? To answer these questions under the constraint that only public interfaces are available, the benchmark synthesises a stratified bilingual corpus of 480 Mandarin Chinese and English sentences with multiple neural voices, applies a main grid of eight degradation conditions together with a babble severity sweep from 20 to -10 dB in eight levels (fourteen distinct degradation conditions in all, plus the clean reference), and measures five recognition systems, three cascade translation paths, and two end-to-end speech translation paths through their public APIs, using only free service quotas. All systems come from one provider, so the study is a single-provider case study whose class-level findings require cross-provider replication before attribution to architecture; this constraint is examined in Section 6 and revisited throughout.

The study yields findings that are relevant to both the science of robust speech processing and the practice of assembling speech services. Recognition services from the three classes differ sharply in where, rather than how fast, they fail under noise: the LLM-based recognisers hold a sub-percent error floor and cross ten percent error at noise levels 2.2 to 2.5 dB lower in Mandarin and 4.4 to 4.8 dB lower in English than the specialised streaming system, with sharper collapse beyond their knee, and below 0 dB they fail by insertion-heavy hallucination that drives character error rates above 100 percent. The end-to-

end translation paths, often assumed to be more robust than cascades because they avoid error propagation, do not inherit robustness automatically: the omni-modal path is the most fragile of the end-to-end paths and, in point estimate under every scoring reference, loses more of its clean quality under severe noise than the LLM-frontend cascades, with a reference-dependent margin, while a streaming end-to-end service is the most resilient path measured. Across up to fifteen operating conditions, the BLEU score of every cascade measured falls linearly with its recognition error rate, an empirical pass-through of 0.44 to 0.54 BLEU points per percentage point whose condition-level uncertainty is wider than its sentence-level confidence intervals and which a three-point re-fit on a second vendor's translation engine reproduces at the same order of magnitude, giving integrators an empirical rule for estimating translation losses from recognition quality alone. Finally, the benchmark surfaces behavioural differences that are invisible in clean-audio evaluations, including two distinct empty-output mechanisms under severe noise, an explicit protocol-level refusal and a silent empty return, and a reverberation fragility of a telephone-band specialist that proves specific to synthetic extreme-DRR room responses rather than to real rooms.

The contribution of this paper is threefold. It provides, to our knowledge, the first systematic measurement of acoustic degradation propagation across recognition and translation stages for commercial black-box speech APIs, covering three product classes of one provider and both cascade and end-to-end translation designs, which we frame as a case study pending cross-provider replication. It validates the central design on real recorded speech under measured impulse responses and a six-level real-babble ladder, bridges the synthetic reverberation condition with the same measured room responses, and extends beyond the primary vendor with an open-weights locally executed recogniser and a second vendor's translation engine in both the reference and the backend role. It introduces a fully reproducible benchmark recipe in which the test material, the degradations, and every model response are controlled, logged, and released, so that the measurements can be repeated as providers update their models.

And it distils the measurements into quantitative design guidance, including noise-failure thresholds per product class, the error pass-through regression for cascades, and a three-reference protocol with one family-disjoint reference that exposes and controls the systematic bias that arises when automatic translation quality is scored against references produced by models related to one of the systems under test.

The remainder of the paper is organised as follows. Section 2 reviews related work on robust ASR, cascade and end-to-end speech translation, and the evaluation of commercial services. Section 3 describes the benchmark design, including the corpus, the degradation grid, the systems under test, and the evaluation protocol. Section 4 presents the results, and Section 5 discusses their implications for practitioners and for future evaluation practice. Limitations are examined in Section 6, and Section 7 concludes.

2. Related work

2.1. Robustness of speech recognition under acoustic degradation

The sensitivity of ASR to noise and channel distortions has been studied since the earliest statistical recognisers [3]. Surveys such as that of Virtanen et al. [4] organise this literature around feature enhancement, model adaptation, and multi-condition training. Controlled evaluation infrastructure has been decisive for progress. The Aurora framework [1] standardised noise types and signal-to-noise ratios for telephone-band recognition, and the CHiME challenge series [2] extended such evaluation to far-field, reverberant, and overlapping-speech conditions. Classical feature transformations such as RASTA filtering [5] targeted the same distortions from the signal-processing side, and synthetic room impulse responses generated with the image method of Allen and Berkley [6] and its modern implementations [7] remain standard tools for controlled reverberation studies. More recently, data augmentation strategies such as SpecAugment [8] have embedded artificial degradation into the training process itself, and weakly supervised training at scale, as exemplified by Whisper [9],

has produced recognisers with markedly improved noise tolerance. The present work inherits the methodological principle of these efforts, namely controlled and reproducible degradation of the input signal, but applies it to black-box services rather than to trainable models, which precludes training-based countermeasures and makes the measurement itself the object of study.

2.2. Cascaded and end-to-end speech translation

The cascade versus end-to-end question has accompanied speech translation since neural systems became practical. Early sequence-to-sequence systems demonstrated that audio-to-text translation without an explicit recognition stage is feasible [10, 11], and Bentivogli et al. [12] compared the two paradigms systematically on IWSLT data, finding that end-to-end systems could match cascades in translation quality while differing in error structure, with cascades inheriting recognition errors and end-to-end systems producing more fluent but occasionally freer output. The survey of Sperber and Paulik [13] synthesises this literature, including work on robustness of end-to-end systems to noisy input. Subsequent research addressed noise robustness of end-to-end ST through self-training and data selection [14]. These studies evaluate models that the researchers train themselves on public corpora. Robustness of the end-to-end component in particular has been treated as a training-data problem, with noise-augmented source audio, self-training on noisy transcripts, and domain adaptation as the standard remedies. That framing presumes control over the model, which is exactly what a service consumer lacks. When the translation component is a fixed API, the robustness question shifts from how to train the model to how to select among available models and architectures under measured channel conditions, and that selection question is what the present benchmark operationalises. The commercially relevant question, namely how the two architectures behave when every stage is a fixed, provider-operated service selected and combined by an integrator, has received far less attention, and the present benchmark is designed specifically for that setting.

2.3. Evaluation of commercial speech services

A third strand of related work evaluates speech services as they are exposed to users. Benchmarks for large audio-language models, such as AIR-Bench [15] for audio understanding and VoiceBench [16] for LLM-based voice assistants, measure knowledge, comprehension, and instruction following, and they establish the black-box, generative-evaluation methodology that the present benchmark extends to the acoustic-robustness axis; neither, however, characterises behaviour under controlled channel degradation, and both stop at the response boundary rather than following errors into downstream tasks. The propagation of recognition errors into translation has a long history in the cascade literature, going back to the statistical framework of Ruiz and Federico [17] for relating ASR error types to downstream MT quality, but that line of work assumes access to the models and corpora involved rather than opaque commercial endpoints. Methodologically, black-box evaluations face constraints that mirror ours: no access to model internals, no control over model versions, and reliance on text normalisation to compare outputs against references. They have, however, largely concentrated on clean or moderately degraded audio, and on transcription accuracy alone. The broader evaluation culture in speech and audio quality research, with its calibrated degradation methodology and emphasis on reproducibility, similarly targets the acoustic layer rather than the linguistic stages downstream, and it is that tradition which the present service-level benchmark follows, adding the service-level measurement that intelligibility metrics cannot substitute for, as the dissociation documented in Section 4.3 shows. The contribution of the present study relative to this strand is twofold. First, the evaluation is extended across the translation boundary, quantifying how recognition differences propagate into translation quality for both cascade and end-to-end architectures. Second, the acoustic degradations are applied under full experimental control, with calibrated noise levels, a synthetic room impulse response, codec and packet-loss emulation, and fixed random seeds, which makes the entire measurement reproducible rather than dependent on whatever happened to be in the test recordings.

2.4. Positioning

In summary, robustness benchmarks address open models but not commercial services, service evaluations address commercial ASR but usually only in clean conditions and only at the recognition boundary, and the cascade-versus-end-to-end literature addresses research models rather than provider-operated systems. CascadeBench combines the three perspectives. It applies the controlled degradation methodology of the robustness literature to the black-box commercial setting, extends the measurement through the translation stage for both architectures, and adds a three-reference scoring protocol, with one reference family-disjoint from every system under test, that quantifies a bias source that any evaluation involving modern generative reference translations must confront. A systematic search of the speech-translation and audio-benchmarking literature, including the benchmarks cited above, found no previous study that measures all of these dimensions simultaneously for the current generation of speech APIs, and none that reports the linear recognition-to-translation error pass-through regression that Section 4 documents; we state this as a finding of our search rather than as a proof of absence.

3. Materials and methods

3.1. Design overview

CascadeBench measures how acoustic channel degradations propagate through commercial speech recognition and translation services that can only be accessed through their public interfaces. The design is shown in Fig. 1. A bilingual corpus of 480 sentences is synthesised to audio with multiple neural voices, degraded by a grid of controlled channel impairments, and submitted to five recognition systems and four speech translation paths. Every stage of the pipeline is scripted with fixed random seeds, and every model response together with its round-trip latency is logged at the sentence level, so that the complete measurement can be re-executed as providers update their services.

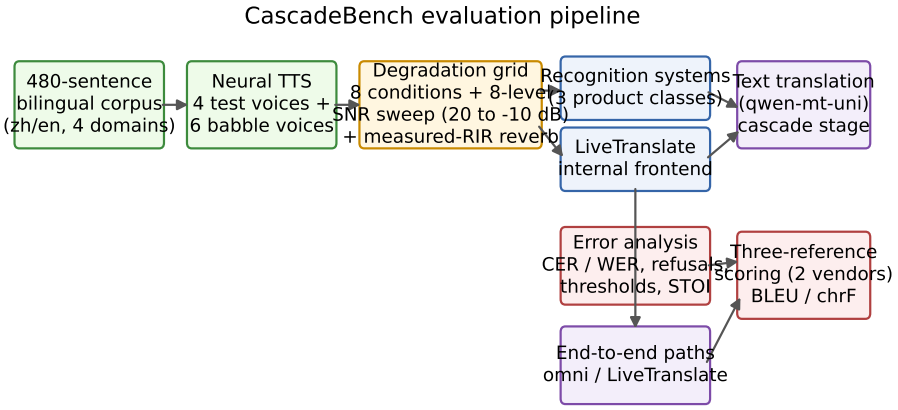


Figure 1: Architecture of the CascadeBench evaluation pipeline. A stratified bilingual corpus is synthesised with multiple neural voices, degraded by a grid of eight acoustic channel conditions plus a noise severity sweep, and processed by recognition systems from three product classes. Cascade systems forward recogniser output to a text translation engine; end-to-end systems translate audio directly.

3.2. Corpus

The corpus contains 240 Mandarin Chinese and 240 English sentences, balanced across four content domains and three length strata, twenty sentences per cell. The domains are broadcast news, everyday conversation, technical and scientific exposition, and a numerals-and-entities stratum rich in dates, quantities, addresses, and named entities, which is known to stress recognition and inverse text normalisation. Short sentences contain 8 to 14 Chinese characters or 5 to 9 English words, medium sentences 15 to 25 characters or 10 to 17 words, and long sentences 26 to 39 characters or 18 to 24 words. The mean length is 19.3 characters for Chinese and 12.0 words for English. Sentences were authored to be self-contained and factually neutral, to avoid brand names and transliterated loanwords whose recognition would depend on provider-specific vocabulary rather than on acoustic processing, and to distribute numerical expressions across date, time, quantity, price, and identifier formats within the numerals domain. All numbers are written out in words in the corpus text, for

example two thousand twenty five, so that the spoken realisation is unambiguous and the reference transcript matches the synthesis input exactly.

Each sentence was synthesised by a neural speech synthesis service (Sambert, Alibaba Cloud Model Studio) using two distinct voices per language, a male and a female voice for Chinese and two native-sounding voices for English, with sentences alternating between the voices so that each voice realises 120 sentences per language. Synthesis produced 16 kHz single-channel PCM audio. The 40 babble noise source tracks described below were synthesised with six additional voices that are disjoint from the four test voices. Using synthesis rather than recorded speech has two methodological advantages. The reference transcript is exact by construction, which removes the transcription noise that limits error floor measurement in recorded corpora, and the acoustic properties of the test material can be regenerated at any time, which decouples the benchmark from the licensing and availability constraints of recorded corpora such as AISHELL-1 [18] or LibriSpeech-derived material [19]. The corresponding limitation, that synthetic speech does not sample the full variability of spontaneous human speech, is examined in Section 6.

3.3. Acoustic degradation grid

Terminology is fixed once here and used throughout. A *degradation family* is one acoustic impairment mechanism; seven families are studied: babble noise, reverberation, narrowbanding, codec compression, packet loss, clipping, and tempo distortion. A *degradation condition* is a family at a specified severity, for example babble at -5 dB; the main grid contributes eight conditions (babble at 10 and 0 dB plus six non-noise conditions) and the severity sweep contributes six further babble levels, fourteen degradation conditions in all. An *operating point* is a cascade evaluation condition, the fourteen degradation conditions plus the clean reference, fifteen in total; the third cascade covers twelve of them.

Eight channel conditions plus the clean control form the main grid. The conditions were selected to span the canonical impairment families that speech services encounter between a talker and a data centre.

Additive babble noise is the most extensively sampled family. Babble was constructed by concatenating randomly selected segments from the 40 disjoint noise-source tracks described above, mixing three such streams, and normalising the mixture, which produces speech-shaped noise with realistic spectral and temporal structure rather than stationary synthetic noise. Speech-shaped interference was chosen over stationary noise because it is the harder and the more ecologically valid case: babble exhibits the temporal modulation and spectral occupancy of competing talkers, masking the target in the same time-frequency regions that carry linguistic information, whereas stationary noise is partly countered by any spectral-subtraction-style front-end that a provider may have deployed internally. The noise was scaled against the active-speech level of each utterance, measured as the mean energy of frames above the 60th energy percentile, to hit target signal-to-noise ratios of 20, 15, 10, 5, 2.5, 0, -5, and -10 dB. Calibration was verified by re-measuring the realised signal-to-noise ratio of the mixed signals against the clean signal; the realised values matched the active-level targets within 0.1 dB on average over the 20 to 0 dB ladder and within 0.6 dB at the -10 dB point. The severity convention deserves emphasis because it is not the only possible one: because the 60th-percentile active-level rule ignores inter-speech pauses, the total-power signal-to-noise ratio of the same signals runs 1.8 to 2.2 dB below the stated active-level values, so a reader comparing against total-power conventions should shift our levels by approximately -2 dB. The 10 dB and 0 dB points enter the main grid, and the full eight-point ladder forms the severity sweep on a stratified 144-sentence subset, 72 sentences per language, sampled evenly from every domain-by-length cell. For the telephone-band recogniser, the degraded signals are generated at 16 kHz first and then downsampled to 8 kHz, so the specialist sees the same noise realisation band-limited to its native input bandwidth rather than noise generated inside its band.

Reverberation was applied by convolving each signal with a synthetic room impulse response of the exponentially decaying noise form [6, 7] with a direct-path impulse and a reverberation time of 0.5 s. We measured the acoustic

properties of the realised impulse responses: Schroeder T_{60} is 0.51 s as designed, the clarity ratio C_{50} is +4.9 dB, and the direct-to-reverberant ratio is −27.6 dB, the last reflecting the exponentially decaying diffuse tail of this synthesis, which concentrates reverberant energy in a spectrally flat late field rather than in discrete early reflections. The condition is therefore best read as diffuse late reverberation at a moderate clarity ratio rather than as an emulation of a specific small room, and its contrast with real recorded impulse responses, whose early-reflection structure colours the spectrum more strongly, is examined in Section 6. Narrowband degradation resamples each signal to 8 kHz and back to 16 kHz with polyphase anti-aliasing filters, which removes the 4 to 8 kHz band exactly as a telephone channel does; the telephone-specialised recogniser receives the 8 kHz signal natively. Codec degradation encodes each signal with the Opus codec [20] at 8 kbit/s mono, in the default voip application mode of the libopus encoder as shipped by ffmpeg 7.1.5, and decodes it back. Measurement of the decoded signals shows that at this bitrate the encoder operates effectively as a narrowband coder: the decoded output retains less than 0.1 percent of clean high-band energy above 4 kHz, so the codec and narrowband conditions are not fully independent stressors of high-frequency content. Packet loss zeroes 15 percent of 20 ms frames chosen uniformly at random, emulating lossy transport without jitter compensation. Clipping applies hard amplitude limiting at 0.3 of full scale, which with a mean input peak of 0.6 clips about 0.6 percent of samples. Tempo distortion resamples the time axis by a factor of 1.15 with pitch preservation using the ffmpeg atempo filter; the spectral centroid is preserved within one percent, no level renormalisation is applied after any degradation, and the peak-normalised output convention follows the synthesis. All random draws use per-sentence and per-condition seeds derived deterministically from a CRC32 digest of the sentence identifier and condition name, and the released audio tree, generated with the same rule, is shipped alongside the logs so that the original measurements are exactly reproducible while fresh regenerations remain deterministic.

Formally, for a clean signal $x[n]$ of length N and a babble realisation $b[n]$,

the mixed signal at level ρ dB is

$$y[n] = x[n] + \sigma\, b[n], \qquad \sigma = \sqrt{\frac{P_{\mathrm{act}}(x)}{10^{\rho/10}\,\overline{P}(b)}}, \tag{1}$$

where $\overline{P}(b)$ is the mean power of the babble and $P_{\mathrm{act}}(x)$ is the active-speech power estimated from frames whose energy exceeds the 60th energy percentile of the utterance, which approximates the ITU-T P.56 active-level convention with a fixed percentile rule. The reverberation impulse response is

$$h[n] = \delta[n] + v[n]\, e^{-6.9\, n/(f_s T_{60})}, \qquad n = 0, \ldots, \lceil f_s T_{60} \rceil, \tag{2}$$

with $v[n]$ unit-variance white noise, $f_s = 16$ kHz, $T_{60} = 0.5$ s, and h normalised to unit energy, and the reverberant signal is the truncated convolution of x with h rescaled to the peak amplitude of x. Packet loss zeroes each independent 20 ms frame with probability $p = 0.15$,

$$y[n] = m_k\, x[n], \quad n \in \text{frame } k, \qquad m_k \sim \text{Bernoulli}(1-p), \tag{3}$$

and clipping applies $y[n] = \mathrm{sign}(x[n])\min(|x[n]|, 0.3)$. Narrowbanding resamples by factors of one half and two with polyphase anti-aliasing filters, codec degradation round-trips the signal through the Opus encoder at 8 kbit/s, and tempo distortion applies a pitch-preserving time-scale modification of factor 1.15.

3.4. Real-speech validation protocol

Because the main corpus is synthesised, we replicated the core of the design on human-recorded speech with human-authored ground-truth transcripts, real measured impulse responses, and real noise, addressing the realism concerns that a purely synthetic pipeline invites. The validation subset contains 150 utterances: 75 Mandarin sentences drawn from the AISHELL-1 test set [18] (11 speakers) and 75 English sentences from the LibriSpeech test-clean set [21] (16 speakers), filtered to 2.5–12 s and level-normalised to the active-speech RMS of the synthetic corpus so that signal-to-noise ratios remain comparable across pipelines. Unlike the synthetic corpus, the reference transcripts are human

transcriptions that never passed through any vendor's synthesiser, which makes the recognition references fully provider-independent.

Three degradation families are applied to each utterance, giving nine conditions in total (clean, real reverberation, six babble SNRs, transport noise), all constructed to be directly comparable to synthetic counterparts. First, real-room reverberation convolves each utterance with one of three measured impulse responses from the MIT Impulse Response Survey [22], selected because their Schroeder-integrated reverberation times (0.495, 0.499, and 0.500 s) match the synthetic $T_{60} = 0.5$ s condition almost exactly while spanning a realistic range of direct-to-reverberant ratios (-0.9, $+1.3$, and $+13.1$ dB); the three rooms are rotated across utterances, and the same three responses, with the same rotation, are additionally convolved with the full 480-sentence synthetic corpus as a bridge condition that isolates impulse-response realism from speech corpus realism. Second, real multi-talker babble is constructed with exactly the same three-stream mixing algorithm as the synthetic babble, but from 40 real recorded utterances (20 AISHELL-1 training speakers and 20 LibriSpeech dev-other speakers, all disjoint from the test speakers), mixed with the target speech at six SNR levels of $+10$, $+5$, $+2.5$, 0, -2.5, and -5 dB, so that the crossing levels of the degradation knee are interpolated between adjacent measured points rather than extrapolated from a three-level tail. Third, real non-speech transport noise, drawn from the train category of the ESC-50 environmental sound corpus [23], is mixed at 0 dB SNR to probe whether the noise-type conclusions transfer beyond speech-shaped interference. Acoustic verification of all degraded signals follows the main protocol: median STOI is 0.77 for real reverberation and 0.71, 0.62, 0.50 for real babble at $+5$, 0, -5 dB.

Four recognition systems (both audio-LLM systems and both streaming specialists) and the Qwen-Audio–cascade and LiveTranslate end-to-end paths are evaluated on all real-speech conditions (clean, real reverb, the six real-babble SNRs, transport noise); the omni system is excluded because its complimentary quota was exhausted. Two external components extend the validation beyond the single vendor of the main grid. The open-weights Whisper small model [9]

is executed locally with the faster-whisper int8 CPU runtime and its default decoding configuration, with an explicit Mandarin initial prompt, on the clean and three hardest babble conditions as a non-vendor recognition baseline; it makes no API calls and its per-call latencies (median 3.9 s, up to 34 s on the hardest conditions) confirm local execution. The GLM large language model of a second vendor (Z.ai), accessed as a hosted API [24, 25], is used in two roles: as a third translation reference, generated from the clean transcripts of both the synthetic and the real corpora, and as an alternative cascade backend, re-translating the Qwen-Audio recognition outputs of the four core real-speech conditions (clean, +5, 0, −5 dB real babble) so that the pass-through relation can be re-fitted on a translation engine disjoint from the vendor of every recognition frontend. Translation references for the validation subset are produced by the oracle protocol: the cascade translation engine applied to the human ground-truth transcripts, so the recognition stage is the only difference between the cascade and its own upper bound. This construction is deliberate for measuring pass-through within the cascade, but it also means that this reference and the cascade share one translation engine: absolute BLEU comparisons against it favour the cascade over any end-to-end path by construction, so the end-to-end comparison in Section 4.4 is drawn in terms of within-path relative losses, which are stable across reference choices, and absolute gaps are interpreted only jointly with the three-reference analysis of Section 4.6. All validation API calls follow the main logging, retry, and scoring infrastructure, and the complete preparation pipeline (dataset selection, impulse-response measurement, babble construction) is seeded and released alongside the main corpus.

3.5. Systems under test

Table 1 lists the systems. The recognition systems of the main grid span three product classes as served by a single commercial provider (Alibaba Cloud Model Studio, Beijing region) during October 2026, accessed through their documented interfaces; the validation components add one open-weights model executed locally and one commercial model of a second vendor, so that vendor

and product class are separated wherever the design allows. The specialised family consists of streaming recognisers in the Paraformer lineage [26], a 16 kHz model and an 8 kHz telephone-band model. The LLM-ASR family consists of Qwen-Audio 3.1 Flash [27, 28] and Qwen3-ASR Flash, audio input models built on large language model backbones. The multimodal family consists of Qwen3.8-Omni Flash, a general-purpose omni-modal foundation model [29, 30] queried with a transcription instruction. The external components are Whisper small [9], the open-weights recognition model, run locally with the faster-whisper int8 CPU runtime, and the GLM large language model of a second vendor (Z.ai) [24], from the GLM foundation-model family [25], used as a reference generator and as an alternative cascade backend. All systems were accessed in their default configurations, since a consumer of a black-box service cannot apply front-end training or adaptation, and the question of interest is precisely how the services behave out of the box. The streaming recognisers were driven by streaming local audio over their realtime WebSocket interface, sending each utterance as 16 kHz or 8 kHz PCM frames at the maximum accepted rate and collecting sentence-final segments. The LLM-ASR and omni systems received base64-encoded whole utterances in their audio-input message format, the omni model together with an instruction prompt for transcription or translation. The streaming translation service was driven over its realtime interface with a session configuration specifying text-only output and the target language, sending raw PCM frames and collecting both the source transcript and the translated text that the service emits, and the text translation engine received source text with explicit source and target language tags. Every call was executed with an automatic retry policy of up to six attempts with exponential backoff on throttling, timeout, and connection failures, and no other tuning parameters were exposed or set. The speech translation systems are the text translation engine qwen-mt-uni [31], used as the second stage of three cascades built on the 16 kHz specialised recogniser, on the Qwen-Audio 3.1 recogniser, and on the Qwen3-ASR recogniser, and two end-to-end paths: the omni foundation model with an audio translation instruction, and a dedicated streaming speech translation ser-

Table 1: Systems under test and their roles in the benchmark. The main grid is operated by a single provider (Alibaba Cloud Model Studio), so product class and vendor are confounded by design there; the validation components separate them: Whisper is an open-weights model executed locally, and GLM is a commercial model of a second vendor. The provider does not disclose the internal architectures, and the class labels follow the product positioning.

System (product class)	Interface	Role	Scope
Paraformer-RT-v2 (specialised)	streaming	recogniser	full grid
Paraformer-RT-8K-v2 (specialised)	streaming	recogniser	zh subset
Qwen-Audio-3.1 (LLM-ASR)	audio API	recogniser	full grid
Qwen3-ASR (LLM-ASR)	audio API	recogniser	full grid
Qwen3.8-Omni (omni)	audio API	recog., E2E	main grid
LiveTranslate frontend	streaming	internal recog.	sweep
qwen-mt-uni (MT)	text API	cascade stage	full corpus
Qwen3.8-Max (LLM)	text API	primary ref.	full corpus
Whisper small (open weights)	local	recog., validation	real subset
GLM (Z.ai, second vendor)	text API	ref. + backend	both corpora

vice (LiveTranslate) that internally combines a streaming frontend recogniser with a translation model. The second-backend validation re-runs the cascade translation stage with the GLM model of the second vendor in place of qwen-mt-uni on the same recognition outputs. The internal frontend of the streaming translation service, which its responses expose as source transcripts, is additionally scored as a seventh recognition system on the severity sweep; the five purchasable recognition systems of Table 1 are therefore complemented by this embedded frontend and by Whisper wherever they are available. All API calls were executed on free-tier quotas with the stop-on-exhaustion option enabled, which both bounds the study at zero monetary cost and documents the practical quota requirements of such a measurement. All systems were accessed with their default decoding configurations and no exposed sampling parameters were overridden, so the repetition analysis of Section 4.10 characterises the services as a consumer actually receives them.

3.6. Evaluation protocol

Recognition accuracy is scored as the character error rate (CER) for Chinese and the word error rate (WER) for English, computed with the standard Levenshtein alignment [32] after a normalisation pipeline that is applied identically to references and hypotheses. The normalisation performs Unicode compatibility folding, case folding, punctuation removal, and, critically, canonicalisation of numeric expressions. Because commercial recognisers differ in their inverse text normalisation conventions, with the same spoken quantity rendered as twenty six, 26, or two six depending on the system, both sides of every comparison are mapped to a canonical digit form by rules that handle Chinese and English cardinal numbers, digit-by-digit readings, time expressions, decimal fractions, and scale multipliers such as wan and yi on the Chinese side and million and billion on the English side, with ordinal expressions masked to a common token. The rule set was developed and verified on the clean condition, where residual normalisation artefacts were inspected sentence by sentence until the clean-condition error rates reported in Section 4 reflect genuine recognition behaviour rather than scoring artefacts. Utterances for which a service returns no transcript at all, a behaviour documented in Section 4, are scored as complete failures with an error rate of one.

Translation quality is scored with corpus-level BLEU [33] using sacreBLEU [34] with the 13a tokeniser for English targets and the character-level tokeniser for Chinese targets, and with chrF [35] as a second string-matching metric; sacreBLEU signatures are fixed in the released scripts (BLEU: effective order 1, case-insensitive, 13a for English and zh character tokenisation for Chinese). Because the corpus is direction-symmetric, BLEU is computed separately for the Chinese-to-English and English-to-Chinese halves and macro-averaged, and the number of scored sentences n is stated for every cell because coverage varies where frontends refuse or sessions drop. Two scoring protocols are distinguished wherever they differ: quality conditional on a non-empty translation, and effective quality in which a refused frontend output is scored as an empty translation. Corpus-level BLEU reacts to empties only through the brevity penalty, which

saturates when hypotheses are longer than references, so the refusal-sensitive comparison in Section 4.5 is anchored on chrF as well. Every hypothesis is scored against three reference sets with deliberately distinct family relationships. The primary reference is produced by Qwen3.8-Max [36], the strongest general-purpose text LLM of the same provider, translating the clean reference transcripts; it shares a vendor, though not a product line, with the translation systems under test. The secondary reference is the oracle cascade output, that is, the cascade translation engine applied to the clean reference transcripts, which is same-engine with every cascade. The tertiary reference is generated by the GLM model of a second vendor [24, 25], translating the same clean transcripts, and is therefore family-disjoint from every system under test; the family relationships of all systems and references are disclosed in Table 1 and revisited in Section 4.6. The three-reference design exposes a bias that Section 4.6 quantifies: automatic scoring against machine-generated references systematically favours systems from the same model family as the reference generator, so conclusions about absolute quality depend on the choice of reference, whereas conclusions about degradation are stable across references. The text normalisation rules were developed by inspecting clean-condition outputs until residual mismatches ceased; a sensitivity analysis in Appendix C re-scores the main grid without numeral canonicalisation and shows that the system ordering is preserved while clean-condition gaps compress. To quantify run-to-run variability of the non-deterministic LLM services, the Qwen-Audio and Qwen3-ASR systems were re-run twice on the 0 dB babble and tempo conditions; condition-level error rates vary by at most 0.9 percentage points across the three runs, although one in five individual transcripts differs between runs (Section 4.10).

Latency is measured as the wall-clock round trip of each API call from dispatch to the complete response, recorded per call. The text translation stage was additionally benchmarked with 120 timed calls on oracle sentences, because the main cascade logs do not carry per-call timing; cascade latency is therefore reported as the sum of the recognition and translation stage means, which is the quantity a consumer experiences when the two stages are invoked sequen-

tially. Streaming systems are fed audio at the maximum rate their interface accepts, so the measurement reflects service-side processing rather than client-side throttling, and it measures whole-utterance turnaround rather than first-partial latency; streaming systems would show substantially lower first-partial latencies than the whole-utterance figures reported here, so the two interface classes should not be compared on absolute latency. All clients ran in one region close to the service endpoint; network round trips are included in all figures.

3.7. Experimental scale and cost

Table 2 reports the exact call accounting of the complete campaign in two columns that separate the two natural counting conventions: *successful calls* counts every logged successful API response, and *unique evaluations* counts distinct (system, condition, sentence) triples after later re-runs have superseded earlier responses. In total 43,046 successful API responses were logged: 26,238 in the original measurement, 5,581 in the extension sweep, 6,675 in the real-speech validation (4,650 for the core six-condition design and 2,025 for extending the real-babble ladder from three to six SNR levels), 3,360 for the measured-impulse-response bridge condition on the synthetic corpus, and 1,192 for the second-vendor tasks (630 GLM reference translations and 562 GLM backend re-translations). Alongside these, 600 Whisper transcriptions were computed locally without API access, and 3,874 failed attempts were logged, of which 3,333 are LiveTranslate websocket sessions that dropped without recovery and 541 are API-level failures (throttling, timeouts, protocol errors) that were retried where the quota allowed. The 875-call difference between the two columns in the original phase consists of re-runs of the clipping, packet-loss, and tempo conditions whose later logs superseded the earlier ones in analysis. Twenty-four logged responses are protocol-level no-words refusals of the Qwen-Audio system under 0 dB babble; they are counted as successful calls, scored as empty transcripts, and tabulated separately in Supplementary Table S4. The 1,544 unique successful LiveTranslate evaluations of the main grid cover 89.5 percent of that system's 1,728 targets, and the extension added 251 of 432 targeted

sessions; all 450 targeted LiveTranslate sessions of the real-speech validation completed successfully; the missing sessions are websocket drops that did not recover within the retry budget, and every reported LiveTranslate figure states its n. The omni system exhausted its complimentary quota before completing its grid: its recognition stage covers the main grid on the 144-sentence sweep subset (119 of 144 sentences at the tempo condition) and its end-to-end translation path covers 83 to 84 of the 144 sweep sentences per condition, which is why omni-based comparisons are restricted accordingly and flagged wherever they appear. The total wall-clock duration of the original data collection was approximately four hours with up to five systems queried in parallel. These figures are reported so that other laboratories can estimate the feasibility of replicating or extending the measurement; they also demonstrate that a multi-system, multi-condition robustness characterisation of commercial speech services is attainable at zero monetary cost with careful quota budgeting, which lowers the barrier for the periodic re-measurement that the field currently lacks.

3.8. Statistical analysis

All aggregate error rates and quality scores are reported with 95 percent bootstrap confidence intervals over sentences, using 1000 resamples. Error rates are sentence-level: the character error rate for Chinese targets and the word error rate for English targets, each computed with the standard Levenshtein alignment, so a Chinese character rate and an English word rate are pooled only as a macro-average of the two language halves, and an insertion-dominated transcript can push a rate above 100 percent, which occurs in the sub-zero-dB regime documented in Section 4.3. Differences between systems within a condition are tested with the Wilcoxon signed-rank test [37] on paired per-sentence errors, with Holm correction [38] within the family of all pairwise comparisons of the same condition and language; because both systems are often simultaneously perfect, the fraction of zero differences is reported alongside (up to 83 percent under reverberation), and the test is then driven by the minority of sentences where the systems differ. Noise severity is charac-

terised three ways: by the piecewise-linear interpolation of the eight-level curve at the ten percent error level, which is model-free and is the quantity tabulated in Table 4, together with a sentence-level bootstrap confidence interval of the crossing; by a four-parameter logistic fit to the same curve, from which the maximum fitted slope in percentage points per dB measures knee sharpness, with the caveat that the fitted knee position of the audio-LLM systems falls between 0 and -5 dB where no observations exist, so the fitted maximum is an interpolation between those levels; and by observed segment slopes between adjacent measured levels, which involve no model at all. Ordinary least squares slopes over sub-ranges are reported only for continuity with common practice, and the text notes how strongly they depend on the chosen range. The relationship between recognition error and translation quality is summarised by a linear pass-through regression with explicit intercept, fitted over up to fifteen operating conditions per cascade. Its uncertainty is reported at two levels deliberately: sentence-level bootstrap intervals, which reflect within-condition sampling noise, and condition-level bootstrap intervals and leave-one-condition-out refits, which reflect how strongly the fitted slope depends on the particular mix of operating points and are the appropriate currency for extrapolating the rule to unseen conditions; the condition-level intervals are substantially wider, and both are reported wherever the rule is used. Signal processing and statistics use SciPy [39], librosa [40], and pystoi for the short-time objective intelligibility analysis of Section 4.3, in both the standard and extended (ESTOI) variants; the tempo condition is excluded from cross-family STOI comparisons because time-scale modification violates the temporal alignment that the STOI family presupposes.

4. Results and analysis

4.1. Clean baseline and the main degradation grid

Table 3 reports sentence-level error rates on the main grid for the five recognition systems, and Fig. 2 visualises the same data split by language. On clean

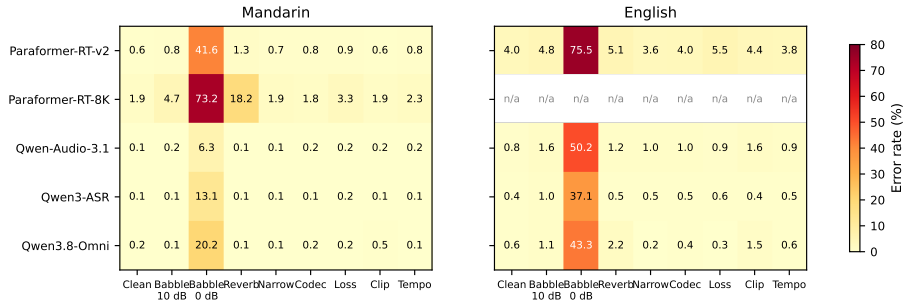


Figure 2: Recognition error rates by system and degradation condition, separated by language. Colour scale is truncated at 80 percent. The 8 kHz specialist is evaluated on the Chinese subset only.

synthetic speech the three foundation-model recognisers are close to their scoring floor, with error rates between 0.3 and 0.5 percent, while the specialised streaming recogniser operates at 0.56 percent on Chinese and 3.99 percent on English, reflecting its primarily Mandarin training. The telephone-band specialist reaches 1.86 percent on its native language. The clean-condition figures establish that the comparisons under degradation are not confounded by differences in clean accuracy of comparable magnitude.

Three regularities stand out in the grid. First, babble noise at 0 dB is by far the most destructive condition for every system, but the magnitude of the collapse separates the product classes sharply. The specialised streaming recogniser reaches a pooled error rate of 58.5 percent and the telephone-band specialist 73.2 percent, whereas the two LLM-ASR systems remain at 25 to 28 percent and the omni model at 31.8 percent. At matched severity, the foundation-model recognisers commit roughly half the error of the specialised systems, and the difference is significant at the sentence level: in Chinese at 0 dB, Qwen-Audio 3.1 errs at 6.3 percent against 41.6 percent for the streaming specialist (Wilcoxon $p < 0.001$, $n = 240$).

Second, the effect of noise is strongly language-asymmetric in a way that differs by class. For the LLM-ASR systems, Chinese recognition at 0 dB remains low, at 6.3 percent for Qwen-Audio 3.1 and 13.1 percent for Qwen3 ASR, while

their English error rates at the same severity reach 50.2 and 37.1 percent. The specialised streaming recogniser suffers heavily in both languages, 41.6 percent in Chinese and 75.5 percent in English. The babble mixture is language-mixed, so the asymmetry cannot be attributed to language-matched interference alone; a plausible partial explanation is that the tonal structure of Mandarin provides complementary cues that survive energetic masking, a hypothesis we state as untested since the benchmark does not isolate tone cues, and one which the general noise-robustness literature does not settle for black-box services [4]. A related measurement caveat is that Chinese is scored at character level and English at word level; the two scales are not directly comparable, and the asymmetry should be read per language rather than across languages.

Third, the non-noise degradations expose narrow specialisation rather than uniform fragility. Narrowbanding to the telephone band costs the LLM-ASR systems 0.3 to 0.6 percentage points of absolute error and the omni model 0.1, whereas the telephone-band specialist shows no measurable home-field advantage over the generalist systems in its own band, where it errs at 1.9 percent against 0.1 to 0.7 percent for the generalists. Reverberation, by contrast, is uniquely expensive for the telephone-band specialist at 18.2 percent, between five and ninety times every other system's score under the same condition, which suggests that its front-end processing is tuned for direct-path telephone speech. Codec compression, packet loss, and tempo distortion at the chosen severities are individually mild for all systems, with the largest single-condition excursion being the packet-loss score of the telephone specialist at 3.3 percent against its clean 1.9. This does not make them harmless: Section 4.5 shows that even sub-point recognition changes propagate measurably into translation quality, and the conditions were deliberately calibrated to deployed-but-aggressive levels rather than to failure-inducing extremes.

Two further behaviours deserve mention because they are invisible in conventional aggregate reporting. The numeral-and-entity domain is five times more error-prone than the other domains for Qwen-Audio 3.1 on clean speech, at 1.25 against 0.25 percent, whereas Qwen3 ASR shows no such penalty, which

indicates that inverse text normalisation rather than acoustics dominates that gap; the normalisation sensitivity analysis of Appendix C supports the same reading. More strikingly, severe noise produces empty final transcripts in three of the five systems, through two distinct mechanisms. Qwen-Audio 3.1 is the only system that refuses at the protocol level, returning an explicit no-words error on 5.0 percent of its 0 dB utterances, concentrated in English at 9.2 percent of that subset; these refusals are scored as complete failures in Table 3. The specialised streaming recognisers fail silently instead: Paraformer-RT-v2 returns an empty transcript on 8.8 percent of its 0 dB utterances without any error signal, and the telephone specialist does so on 0.8 percent. Qwen3-ASR and the omni model always emit a hypothesis, however wrong. For a downstream consumer the two mechanisms differ in detectability: an explicit refusal can be caught and retried, whereas a silent empty return is indistinguishable from a genuine silent segment, and this distinction is the kind of behavioural divergence that only black-box evaluation at controlled severity can surface. The full per-system, per-condition refusal rates, including the sub-zero regime where they rise above 40 percent, are tabulated in Supplementary Table S4.

The remaining conditions merit condition-specific remarks because their effects are small in aggregate but structured in detail. Reverberation with a reverberation time of 0.5 s raises the specialised streaming recogniser to 3.2 percent pooled, but the effect concentrates in English, where it reaches 5.1 percent against 1.3 percent in Chinese, and the paired difference against the LLM-ASR systems remains significant in both languages (Wilcoxon $p < 0.01$ in Chinese, $p < 0.001$ in English; 72 to 83 percent of paired differences are zero under reverberation, so the significance is carried by the minority of sentences where the systems differ). The omni model shows an idiosyncratic signature under reverberation and clipping, at 1.1 and 1.0 percent, roughly triple its clean error rate, which suggests that its audio encoder is more sensitive to spectral colouration and non-linear distortion than the dedicated recognisers even though it holds up well under noise. Codec compression at 8 kbit/s leaves every system within 0.3 percentage points of its clean score, which is expected in retrospect because

measurement of the decoded audio shows the 8 kbit/s Opus stream to be effectively narrowband, so the codec condition largely re-tests the narrowband condition at a different distortion profile. Packet loss at 15 percent and clipping at the 0.3 threshold behave as mild aggregate stressors, and tempo distortion at 1.15 times is fully transparent to every system, which is expected because time-scale modification preserves the spectral envelope that recognisers rely on. The practical reading of these patterns is that a service integrator who cannot control the acoustic environment should prioritise noise mitigation far above codec or transport tuning, since the latter contribute almost nothing to the error budget at realistic operating points.

4.2. Stratified analysis by domain

Because the corpus is balanced by construction, error rates can be decomposed by content domain without re-weighting. On clean audio the technical domain is the most error-prone for the specialised streaming recogniser at 3.3 percent, consistent with its terminological density, while the conversation domain is the easiest for every system, below 0.1 percent for the Qwen-Audio recogniser. The numerals-and-entities domain shows the clearest family split on clean speech: the two LLM-ASR systems differ by an order of magnitude on it, at 1.2 percent for Qwen-Audio against 0.15 percent for Qwen3-ASR, which as noted in Section 4.1 points to inverse text normalisation rather than acoustics as the separating factor. Under 0 dB noise the domain differences compress for every system: the spread across domains narrows to about nine points for the Qwen-Audio recogniser and to twelve points for the specialised streaming recogniser, and the ordering changes, with the technical domain becoming the most difficult for both. The compression indicates that under severe energetic masking the bottleneck shifts from lexical and normalisation knowledge, which distinguishes systems on clean speech, to the acoustic recovery problem, which is common to all systems and dominates the error budget.

4.3. Scaling with noise severity

Fig. 3 shows error rate as a function of signal-to-noise ratio on the 144-sentence severity sweep, now spanning 20 to −10 dB in eight levels, and Table 4 reports the fitted severity characteristics. The dominant shape is a knee rather than a line: every system holds a nearly flat floor above roughly 10 dB and transitions to near-total failure within a few dB below its knee point, so a single percentage-points-per-dB slope is not a stable summary of these curves. The slope fitted over the original 20 to 0 dB range, for example, moves from −0.26 to −4.69 points per dB for the Qwen-Audio system in Mandarin when the fit is extended to −10 dB, a range dependence that reflects the flat-then-cliff structure rather than any measurement change. We therefore characterise severity response by the signal-to-noise ratio at which the fitted error rate crosses ten percent, with sentence-level bootstrap confidence intervals, together with the maximum fitted slope as a measure of knee sharpness.

The crossing thresholds separate the classes more informatively than slopes. In Mandarin the specialised streaming system crosses ten percent error at 2.48 dB (95% CI 2.26 to 3.23), the telephone specialist at 8.82 dB (8.17 to 9.31), and the two LLM-ASR systems at −0.06 and 0.29 dB, a tolerance advantage of 2.19 to 2.54 dB over the streaming specialist and 8.53 to 8.88 dB over the telephone specialist. In English the ordering is the same but the margins widen: the streaming system crosses at 6.79 dB (4.93 to 7.91) against 2.38 dB (2.02 to 3.38) for Qwen-Audio and 2.02 dB (1.73 to 2.31) for Qwen3-ASR, a 4.41 to 4.77 dB advantage. The LLM-ASR curves also have the sharpest knees: their fitted maximum slopes reach 21 to 36 percentage points per dB against 10 to 21 for the specialised systems. Two honesty notes attach to these fitted maxima. First, the fitted knee positions of the two LLM-ASR systems in Mandarin lie at −2.7 and −3.6 dB, inside the 0 to −5 dB interval where the sweep has no observation level, so the fitted peak is a model interpolation rather than a measured quantity. Second, the observed segment slopes, which involve no model, tell a compatible but more cautious story: the steepest measured segments are 23.7 percentage points per dB for Qwen-Audio and 18.7 for Qwen3-ASR (both Man-

darin, between 0 and −5 dB), against 13.0 for the streaming specialist (between 2.5 and 0 dB). The LLM-ASR systems are therefore better read as flat-floor-then-cliff systems than as gradually degrading ones, with the cliff confirmed by observation but its precise sharpness extrapolated. The mild-noise regime above 10 dB is effectively free of robustness differences in both languages: every system stays within one point of its clean score. Between 10 and 5 dB the specialised systems begin to move first, and by 5 dB the class ordering is established in both languages. The practical implication for procurement testing is that babble probes near each system's knee, in the 0 to 5 dB window here, separate the classes most sharply, whereas a single 5 dB probe ranks systems reliably at the top of this severity range but the LLM-ASR systems are still within noise of each other at that level.

The three levels below 0 dB reveal a regime that the upper ladder cannot see. All systems approach total failure, but the LLM-ASR systems fail distinctively: their Chinese error rates exceed 100 percent, reaching 127 percent for Qwen-Audio and 143 percent for Qwen3-ASR at −10 dB, which is only possible when inserted characters outnumber the reference length. Inspection of the transcripts shows long repetition loops and fluent but unrelated text, that is, insertion-heavy hallucination rather than garbled acoustics, a failure mode that a wrong-but-bounded view of error rates cannot represent and one that is particularly dangerous downstream because a hallucinated transcript is grammatical text that a translation stage will happily translate. In the same regime the streaming systems simply return fragments or silence, and their refusal rates rise to 42 to 58 percent (Supplementary Table S4). No system transcribes anything reliably below 0 dB, and the appropriate engineering response in that regime is capture improvement rather than model selection.

The severity sweep also anchors the benchmark in classical signal quality. Short-time objective intelligibility (STOI) computed against the clean signals falls monotonically with noise level, from 0.97 at 20 dB through 0.67 at 0 dB to 0.40 at −10 dB, and within the babble family the condition-level correlation between STOI and recognition error is strong. Across degradation families,

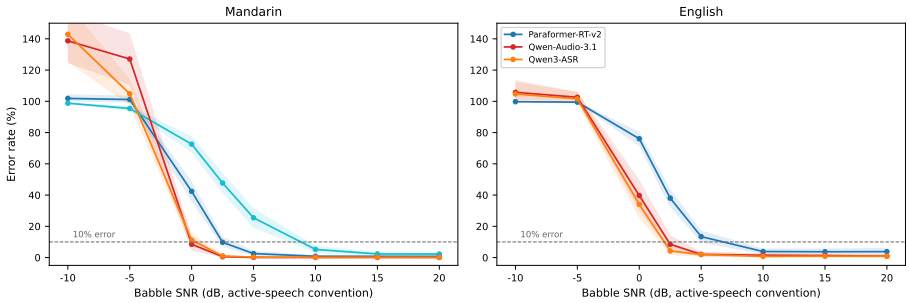


Figure 3: Recognition error rate against babble noise signal-to-noise ratio for four systems and both languages, with bootstrap 95 percent confidence bands, on the 144-sentence severity sweep (72 sentences per language per point). Dashed horizontal line marks the ten percent error level used for the knee thresholds of Table 4.

however, STOI and error dissociate sharply: reverberation at a measured STOI of 0.63 leaves every generalist system below 3.2 percent error, while 0 dB babble at a comparable STOI of 0.67 produces 25 to 59 percent error. The extended variant ESTOI, which is more sensitive to fluctuating maskers, sharpens the noise ladder (0.93 at 20 dB, 0.43 at 0 dB, 0.14 at -10 dB) but reproduces the same dissociation, with reverberation at 0.43 comparable to 0 dB babble at 0.43. The tempo condition is deliberately excluded from this comparison: time-scale modification destroys the temporal alignment that both STOI and ESTOI presuppose, so any number they assign to it is a property of the metric's assumption rather than of the signal, and intelligibility-oriented signal metrics therefore do not predict black-box recognition error across degradation families. Service-level benchmarks of this kind measure something the signal-quality literature cannot substitute for, with the corollary that an integrator cannot infer API robustness from instrumental quality scores alone.

4.4. Validation on real recorded speech

The real-speech protocol of Section 3.4 replicates the central regularities of the synthetic pipeline while calibrating two of its absolute levels. On clean real recordings all four systems remain usable, with error rates of 0.8–1.2% for the audio-LLM systems, 3.6–4.1% for the 16 kHz streaming specialist, and 5.1% for

the telephone-band specialist; these are within a few points of their synthetic clean floors, confirming that near-clean performance measured on synthetic speech transfers to human recordings of read speech. The measured-impulse-response bridge condition locates the reverberation gap precisely, because it applies the same three survey room responses to the same synthetic corpus as the synthetic exponentially decaying condition. The audio-LLM systems are insensitive to the distinction (Qwen-Audio 0.2/1.1% Mandarin/English under measured responses against 0.2/1.2% under the synthetic response, Qwen3-ASR 0.2/0.5 against 0.1/0.5), the 16 kHz specialist improves from 1.3/5.1 to 0.7/3.7 percent, and the telephone-band specialist improves from 18.2 to 3.4 percent Mandarin CER; at the cascade level the measured responses gain 0.5 to 1.4 BLEU over the synthetic response under all three references (Qwen-Audio cascade 53.8 against 52.4 against the primary reference on the matched sweep subset). The explanation is acoustic: the synthetic response's direct-to-reverberant ratio of -27.6 dB is far below the survey median of $+7.3$ dB for everyday spaces at conversation distance, so the synthetic condition is best read as a stress case at the reverberant extreme rather than a typical room, and the telephone-band specialist, whose narrowband input magnifies reverberant smearing, is the system most affected by that choice.

Real multi-talker babble reproduces the knee-and-collapse structure of Fig. 3 with a harder noise floor (Fig. 4), and the six-level ladder, tabulated in full in Appendix D, measures the crossings directly rather than interpolating them from a three-level tail. Qwen-Audio crosses ten percent Mandarin error at 2.8 dB (English 2.3 dB), Qwen3-ASR at 7.0 dB (English 3.6 dB), while both streaming specialists already exceed ten percent Mandarin error at $+10$ dB, the mildest level tested, so their crossing levels lie outside the measured range. Qwen3-ASR is therefore better than both specialists at the milder tested levels but clearly worse than Qwen-Audio, and its position against the specialists at harsher levels is language-dependent: in English it stays below the 16 kHz specialist at every tested level (64.1 against 86.1 percent at -5 dB), whereas in Mandarin it is already behind the telephone-band specialist at $+2.5$ dB (62.3 against 60.2

percent) and behind both specialists from 0 dB downward. This level- and language-dependent structure is exactly what a three-level babble ladder cannot resolve. The class-level advantage on real babble is larger than on synthetic babble: Qwen-Audio holds sub-15% Mandarin error at +5 dB (4.7%) where both specialists already err at 40.1 and 48.6 percent, and the LLM noise-tolerance advantage, 2.2 to 2.5 dB over the streaming specialist on synthetic babble, exceeds seven dB here. The advantage is nonetheless confined to the near-knee regime: at 0 dB Qwen-Audio errs at 46.9% (Mandarin) against 102.6% for the 16 kHz specialist, and at −5 dB every system except the telephone-band specialist exceeds 130 percent, so the correct summary is not that the LLM class advantage widens without bound, but that it holds in the near-knee and low-error regime and disappears, for one of the two LLM systems outright reverses, once error rates pass the ten-percent level into the hallucination regime.

The open-weights Whisper small model, executed locally on the same real-speech conditions, clarifies which of these regularities are engine-specific. On clean audio it is usable (10.1% Mandarin CER, 3.6% English WER), but at +5 dB real babble it errs at 78.7 percent Mandarin against 19.3 percent English, so its ten-percent crossing level lies above the mildest tested babble SNR and its knee position cannot be localised within the measured range; the data neither confirm nor refute the knee structure for this model. What the Whisper outputs do show is the failure structure. In the high-error region its Mandarin transcripts develop the same insertion-loop hallucinations as the provider's LLM recognisers – insertions outnumber deletions by 244 to 83 in the +5 dB Mandarin error mass, and the canonical repetition-loop transcript, a single two-character word repeated to length, appears verbatim – while its English outputs remain deletion-dominated, and its silent-empty rate rises from 5.3 percent at +5 dB to 25.3 percent at −5 dB in Mandarin. The language asymmetry that replicates across engines is therefore one of failure structure – Mandarin insertion-dominated against English deletion-dominated – and not one of noise tolerance: the provider's LLM recognisers are more noise-tolerant in Mandarin than in English on the synthetic grid, whereas Whisper small is dominated by Man-

darin errors already at +5 dB, so language-specific tolerance levels are engine properties rather than a general law.

The −5 dB hallucination regime replicates and sharpens the synthetic finding, and the error-type decomposition locates it precisely. Mean Mandarin CER exceeds 100% for every provider system (150.0% for Qwen-Audio, 207.6% for Qwen3-ASR, 164.4% for the 16 kHz specialist), 53–79% of LLM utterances exceed 100% CER individually, and the Qwen-Audio refusal rate rises to 29.3% (Mandarin) and 18.7% (English), closely matching the synthetic-condition refusal behaviour. Decomposing the alignment errors shows that the insertion dominance is a Mandarin phenomenon: at −5 dB real babble, insertions account for 45% of Qwen-Audio's and 56% of Qwen3-ASR's Mandarin error mass against 1.6% for the telephone specialist, whereas in English every system is deletion-dominated (insertion fractions 0.9 to 12.4%), consistent with the synthetic sub-zero pattern where the LLM systems' Mandarin insertion fractions reach 29–34% but their English fractions stay below 6%. Real non-speech transport noise at 0 dB is degradative but far gentler than real babble (10.9% CER for Qwen-Audio against 46.9%), confirming that speech-shaped interference is the harder stressor for these systems regardless of whether the babble is synthetic or recorded.

The cascade conclusions also transfer, with one scoring caveat that the validation design itself surfaces. Cascade BLEU on real speech falls monotonically from 69.2/68.9 (Mandarin/English, clean) through 63.7/68.6 at +5 dB and 43.7/47.8 at 0 dB to 6.9/7.6 at −5 dB, and regressing BLEU loss on recognition error rate across the nine conditions gives an empirical pass-through of 0.43 (Mandarin, $r = 0.994$) and 0.72 (English, $r = 0.999$), bracketing the synthetic-grid values of 0.44 to 0.54; the direction difference between the two slopes is itself part of the finding, and the condition-level intervals, not the sentence-level ones, are the honest uncertainty of a relation fitted over nine operating points. The end-to-end comparison, by contrast, must be read through the reference lens established in Section 4.6: the validation reference is the cascade's own engine applied to human transcripts, so it is same-engine with the cascade

and foreign to LiveTranslate, and the absolute gap of 27 to 28 BLEU on clean speech (69.1 against 42.0) is therefore not a translation-quality measurement. Two observations follow, one reference-independent and one quantifying the reference sensitivity itself. First, the mechanism of the bias is visible directly: on clean real speech the cascade's ASR frontend reproduces the human transcript exactly on 133 of 150 sentences, and the engine reproduces its own reference exactly (self-BLEU 100), so the cascade's 31-point distance from the reference on the remaining 17 sentences shows how strongly the engine's word choice responds to punctuation and inverse-text-normalisation differences in its input, which BLEU then penalises at full weight. Second, the relative-loss comparison between the two paths is itself reference-sensitive, and a paired sentence-level bootstrap (1000 resamples over the 150 sentences) keeps every difference compatible with zero: from clean to 0 dB real babble under the same-engine reference, LiveTranslate loses 41.6 percent of its own clean BLEU against 33.7 percent for the cascade (difference +7.9 points, 95% CI −1.5 to +16.7), while under the family-disjoint GLM reference the two paths are statistically indistinguishable, losing 32.6 and 32.7 percent respectively (difference −0.1, CI −9.0 to +9.0; Supplementary Table S12). The real-speech data are therefore consistent with the synthetic-grid finding that end-to-end paths are not automatically more robust – the streaming end-to-end path is the more fragile one in point estimate under the same-engine reference – but they do not establish that fragility at the sentence level, and the absolute BLEU ordering of the two paths on this reference is reported only as a same-engine artefact, in line with the three-reference analysis of Section 4.6.

4.5. Cascade versus end-to-end translation under degradation

Fig. 5 compares the speech translation paths across the babble sweep, and Table 5 reports the corresponding scores with all three references, bootstrap confidence intervals, sample sizes, and the refusal-inclusive chrF. On clean audio the three cascades lose almost nothing relative to their translation oracle, the text translation engine applied to perfect transcripts, which scores 55.5 BLEU

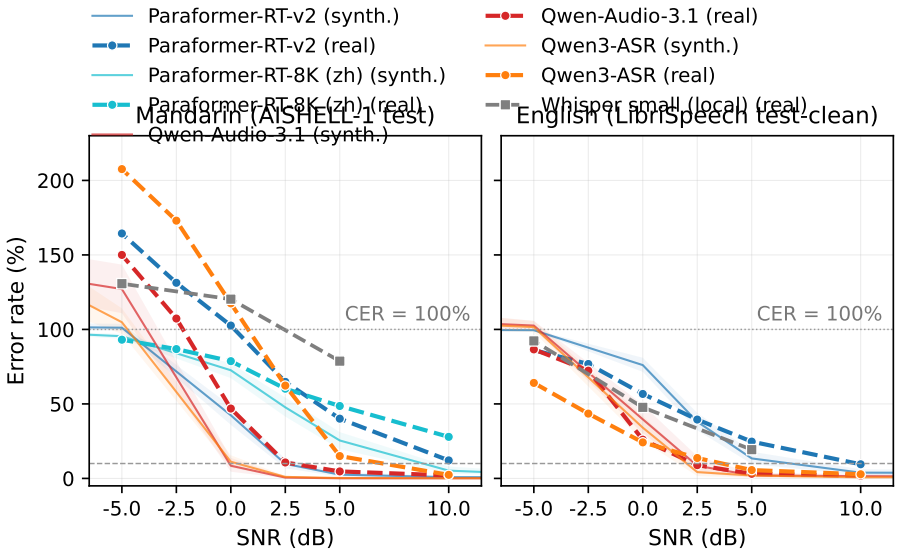


Figure 4: Recognition error rate versus SNR under synthetic babble (thin solid lines with 95% bootstrap bands, 8-point sweep) and real multi-talker babble (thick dashed lines with markers, six SNR levels from +10 to −5 dB) on real recorded speech; the grey dashed series with square markers is the locally executed Whisper small model on its four tested conditions. Left: Mandarin (AISHELL-1 test, telephone-band specialist Mandarin only); right: English (LibriSpeech test-clean). Real babble is the harder stressor at every SNR, but the system ordering, the knee structure, and the sub-0 dB hallucination regime (CER above 100%, dotted line) all replicate; Whisper small's Mandarin error already exceeds the knee at +5 dB, the mildest level it was run at.

against the primary reference. The cascade built on Qwen-Audio 3.1 starts at 53.9 BLEU (n=144), the cascade built on the streaming specialist at 51.5, the cascade built on Qwen3-ASR at 53.2, and the end-to-end omni path scores highest at 68.1 (n=84) with the streaming LiveTranslate service at 59.8 (n=144). Under degradation the ordering and the magnitudes change dramatically. At 0 dB the Qwen-Audio cascade drops 9.8 BLEU points, the streaming-specialist cascade collapses by 35.7 points to 15.8 BLEU, the Qwen3-ASR cascade drops 10.2 to 43.0, the omni end-to-end path loses 22.5 points, and the streaming translation service loses 9.3 points from a lower starting level. The sentence-level confidence intervals of Table 5 are wide relative to these movements, so the

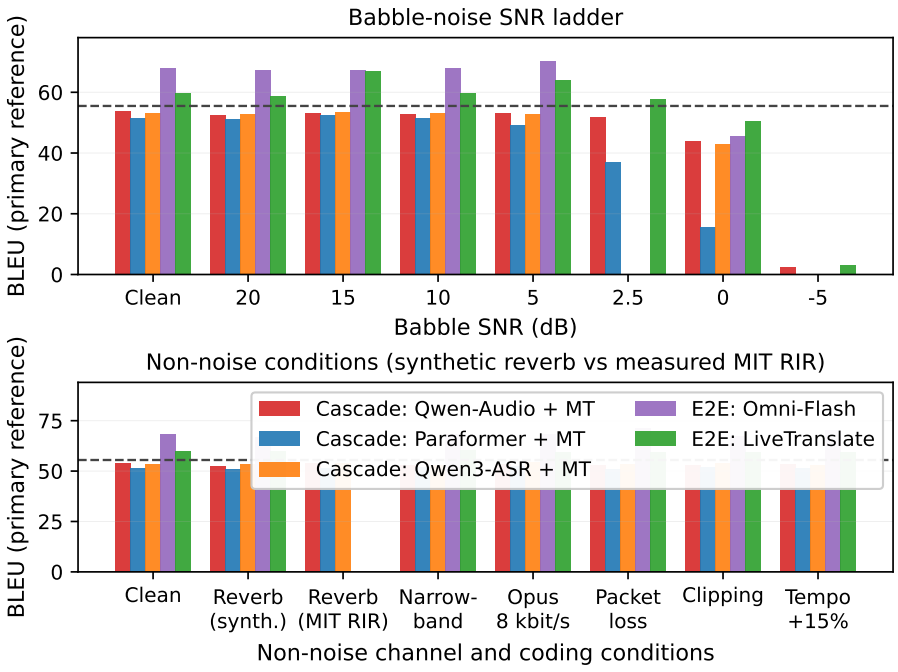


Figure 5: Translation quality of the three cascades and the two end-to-end speech translation paths. Upper panel: the babble SNR ladder. Lower panel: the non-noise channel and coding conditions, including the synthetic-reverb condition and the measured MIT-RIR bridge condition on the matched sweep subset (the two end-to-end paths were not run under the measured RIRs, so their reverb bars are absent). All bars are scored against the same-vendor LLM reference (Section 4.6); the dashed line is the text-translation oracle on clean transcripts. Per-condition confidence intervals are tabulated in Table 5 and in the supplementary material.

0 dB collapses are unambiguous while adjacent-system differences of one to two points are not. The refusal-inclusive chrF tells the same story with the empty outputs counted: the Qwen-Audio cascade falls from 65.2 to 53.7 chrF at 0 dB once its eight refused sweep sentences are scored as empty translations, a drop that corpus BLEU would understate because its brevity penalty saturates when hypotheses are longer than references, which is why both metrics are reported.

Three conclusions follow. The first concerns the frontend hypothesis. The translation quality of a cascade under acoustic degradation is governed largely by the robustness of its recognition stage: swapping the frontend from the

streaming specialist to the LLM recogniser at 0 dB is worth 28 BLEU points (44.1 against 15.8) on the same backend, and the third cascade, built on Qwen3-ASR, tracks the Qwen-Audio cascade within 1.1 BLEU points (mean gap 0.5, maximum at 0 dB) across the twelve shared conditions, which shows that the pattern generalises across frontends of the same class. What the main grid cannot separate is the contribution of the translation engine, since every cascade shares one backend; the frontend comparison is within-backend and valid, and the second-backend validation of Section 4.8 varies the engine on the real-speech subset to test how far the pass-through relation travels. The second qualifies the common expectation that end-to-end systems avoid cascade error propagation and are therefore more robust. Among the end-to-end paths the omni model is clearly the more fragile one at 0 dB, and in point estimate it also loses a larger fraction of its clean quality than the LLM-frontend cascades under every reference (33 against 18–19 percent primary, 30 against 17–23 GLM); the margin, however, depends on the reference, shrinking to under one point against the Qwen3-ASR cascade under the oracle reference (29.0 against 29.5 percent), and on the sentences common to the paths the paired bootstrap of the relative-loss difference crosses zero under the oracle reference (+11.7 points, 95% CI −5.6 to +25.8 against the Qwen-Audio cascade; +4.4, CI −9.2 to +18.4 against the Qwen3-ASR cascade) while remaining positive under the primary and GLM references. The omni path's apparent absolute lead over the cascades is, moreover, an artefact of two confounds: its clean score of 68.1 is measured on a quota-limited subset of 84 sentences and carries reference affinity (Section 4.6), and on the sentences common to both paths the 0 dB absolute comparison is statistically indistinguishable under the primary and GLM references (Qwen-Audio cascade 47.1 against omni 45.5, paired CI −11.3 to +5.5 primary; −10.0 to +6.3 GLM), separating in the cascade's favour only under the oracle reference, whose same-engine affinity the reference analysis predicts. The defensible statement is therefore about degradation behaviour, not absolute ranking: the omni path is the most fragile of the end-to-end paths and, in point estimate, degrades more than the LLM-frontend cascades at 0 dB, with a reference-dependent margin

whose paired uncertainty crosses zero under the oracle reference. The streaming LiveTranslate service is the most degradation-resilient of the paths measured, losing 16 to 21 percent at 0 dB depending on the reference, which is consistent with the behaviour of its internal frontend recogniser documented in Section 4.9. End-to-end designs do not inherit robustness automatically; they inherit the robustness of whatever acoustic frontend their provider trained internally, which in this measurement is better than the specialised streaming ASR but not automatically better than the best cascade.

The condition-wise structure of Fig. 5 adds three nuances to this picture. First, apart from the 0 dB point the cascade built on the LLM recogniser is essentially flat across the entire grid, moving within one BLEU point of its clean score under every other condition, which mirrors the recognition flatness of Table 3 and confirms that the non-noise degradations are, at deployed severities, engineering noise rather than failure modes for a robust frontend. Second, the streaming translation service shows the same flatness with a single exception: its narrowband score of 60.1 sits marginally above its clean score, and its packet-loss and tempo scores are likewise within half a point of clean, so its 0 dB loss is the only substantive degradation it suffers. Third, the omni end-to-end path exhibits a small but consistent anomaly: its packet-loss and tempo scores, at 71.1 and 70.0, exceed its clean score of 68.1. Inspection of the outputs suggests that under these mild distortions the model produces slightly more concise translations that happen to align better with the reference style, rather than genuinely better translations; the effect is within the reference-sensitivity margin documented in Section 4.6 and is a further argument for treating small absolute differences between heterogeneous systems with caution.

Decomposing the 0 dB collapse by direction sharpens the frontend story. The specialist cascade loses 25.9 BLEU points in the Chinese-to-English direction but 45.6 points in the English-to-Chinese direction, falling to 8.7 BLEU, because its English recognition, at 75.5 percent error, destroys the source representation before the translation stage operates. The LLM-frontend cascade degrades far less but still asymmetrically, losing 4.7 points in the Chinese-to-English direction

and 15.0 in the other, a 3.2 to one ratio, and the omni path shows the same skew at 19.4 against 25.7. The streaming translation service loses almost nothing in the Chinese-to-English direction at 0 dB, at 56.0 against 57.3 clean, but 18.3 points in the other direction, which parallels the language asymmetry of its internal frontend. No path degrades symmetrically, so direction-symmetric deployment claims should be checked per direction rather than pooled.

4.6. Reference sensitivity of translation scoring

Table 5 reports relative losses against all three references: the primary reference produced by the provider's strongest text LLM, the oracle reference produced by the cascade's own translation engine, and the GLM reference produced by a second vendor. The three references disagree sharply about absolute quality and can invert apparent orderings. Against the primary reference the omni end-to-end path leads on clean audio at 68.1 BLEU, exceeding even the text-translation oracle at 55.5; against the oracle reference the same path scores 48.5 while the Qwen-Audio cascade scores 89.6, because the cascade shares its translation engine with the reference generator; and against the GLM reference the omni path's clean lead shrinks to 60.6, since the primary reference shares a vendor with the omni model while GLM does not. The same triangle repeats at 0 dB: the omni path and the Qwen-Audio cascade are statistically indistinguishable against the primary and GLM references but 43 points apart against the oracle reference. The pattern is exactly what same-family affinity predicts, and the GLM reference, which is family-disjoint from every system under test, is the one against which no system enjoys affinity by construction.

Two regularities nevertheless survive all three references. First, degradation losses are computed within a system against a fixed reference, so the percentage losses are comparable across references even where absolute scores are not. Second, the ordering of relative losses at 0 dB is stable across references with a single exception: under the primary and GLM references the ordering is LiveTranslate, Qwen-Audio cascade, Qwen3-ASR cascade, omni, Paraformer cascade (16–21, 17–18, 19–23, 29–33, 69–80 percent), while under the oracle ref-

erence the omni path (29.0) and the Qwen3-ASR cascade (29.5) swap adjacent positions, differing by less than one point. The honest summary is therefore that the relative-loss ranking is stable across the three references except for a sub-point inversion between the omni path and the Qwen3-ASR cascade under the oracle reference, an inversion small enough that no conclusion in this paper rests on their separation. We recommend that black-box translation evaluations score against at least two references generated by different model families from each other and from every system under test, state the family relationships explicitly, and treat cross-family absolute comparisons as provisional until human-reference anchored.

4.7. Error pass-through from recognition to translation

Fig. 6 relates the BLEU loss of the three cascades to their mean recognition error rate across all operating conditions, fifteen per cascade for the two original frontends and twelve for the Qwen3-ASR cascade, spanning recognition error rates from 0.3 to over 100 percent. The error metric on the horizontal axis is the sweep-subset mean recognition error with refusals counted as complete failures, matching the protocol of Table 3, and the vertical axis is the BLEU loss from each cascade's clean score against the primary reference. The relationship is strongly linear across this range: the Pearson correlation is 1.00 for the Qwen-Audio cascade, 0.995 for the specialist cascade, and 0.994 for the Qwen3-ASR cascade. Three cautions, however, separate this empirical relation from a law, and each is quantified rather than asserted. First, the correlation is concentrated in a few high-error conditions: the points cluster near the origin for the nine mild conditions, so the line is anchored chiefly by the four to five collapsed conditions, which is why leave-one-condition-out refits are stable for the first two cascades (slopes move by less than 0.03) but not for the third, where dropping the 0 dB condition moves the Qwen3-ASR slope from 0.47 to 0.74. Second, the sentence-level bootstrap intervals understate the design uncertainty: resampling the operating conditions themselves gives 95 percent intervals of 0.23 to 0.45 for the Qwen-Audio cascade, 0.52 to 0.66 for the specialist cascade, and 0.38 to 3.9

(heavily right-tailed, median 0.47) for the Qwen3-ASR cascade, so the condition mix, not the sentence sample, is the dominant source of uncertainty in the slope. Third, the slope is not constant across the error range: excluding the conditions above 40 percent recognition error moves the slope from 0.44 to 0.37 for the Qwen-Audio cascade and from 0.54 to 0.68 for the specialist cascade, and the linear form cannot hold without bound on the high side, because BLEU loss saturates at the clean score while recognition error can exceed 100 percent through insertions. Within these bounds the fitted relation, with explicit intercepts of $+0.6$ and -1.0 BLEU points (both statistically indistinguishable from zero, and small enough that the through-origin reading changes no prediction by more than a point), gives slopes of 0.44 BLEU points per percentage point of recognition error for the Qwen-Audio cascade (sentence-level bootstrap 95% CI 0.41 to 0.48), 0.54 for the specialist cascade (0.51 to 0.58), and 0.47 for the Qwen3-ASR cascade (0.35 to 0.60). The steeper slope for the specialised frontend reflects the composition of its errors under severe noise, which include whole-utterance failures where the translation engine receives little or no usable signal, whereas the LLM recogniser's residual errors under moderate degradation are concentrated in individual words that the translation engine can often route around. A service integrator can therefore use the relation as an empirical rule of thumb, valid for error rates up to roughly 60 percent and for translation engines of the class tested here: every percentage point of recognition error costs roughly half a BLEU point downstream, with the caveat that the cost drifts toward two thirds of a point for error profiles dominated by whole-utterance failures and toward two fifths for error profiles dominated by isolated word errors, that the honest uncertainty of the coefficient is the condition-level interval rather than the sentence-level one, and that the main-grid estimate is calibrated on one translation engine, although the second-backend validation of Section 4.8 reproduces the relation on a disjoint engine with the recognition frontend held fixed.

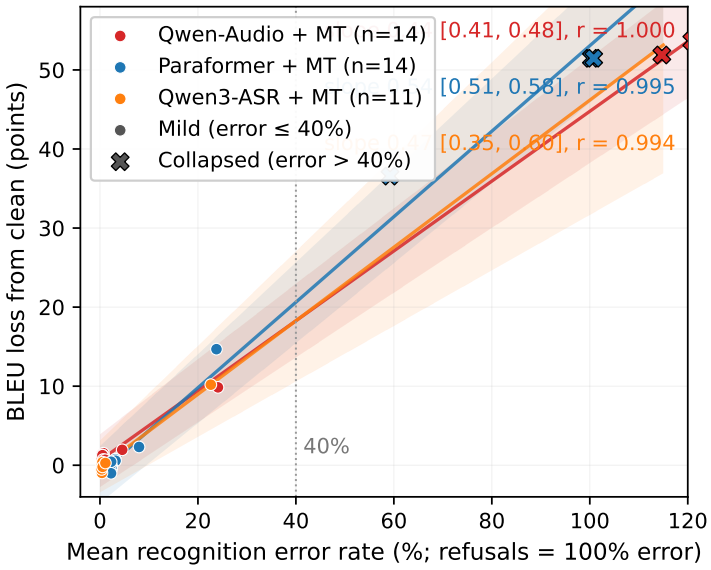


Figure 6: Translation quality loss against mean recognition error rate for three cascades over fifteen operating conditions for the two original frontends and twelve for the Qwen3-ASR frontend, with fitted linear pass-through (with intercept) and 95% confidence bands. Recognition error is the sweep-subset mean with refusals counted as complete failures; sub-zero-dB conditions extend the range beyond 100 percent through insertion errors. Mild conditions (circles) cluster near the origin and collapsed conditions (crosses, recognition error above 40 percent) anchor the fit. The clean condition has zero loss by construction and is excluded from the fit, so the regression in the legend uses $n = 14$, 14, and 11 points per cascade.

4.8. Pass-through under a second translation backend

Because every cascade in the main grid shares one translation engine, the pass-through relation of Fig. 6 could in principle be an artefact of that engine. The second-backend validation re-translates the Qwen-Audio recognition outputs of the four core real-speech conditions with the GLM model of a second vendor (Z.ai), a translation engine disjoint from the provider of every recognition frontend, keeping the recognition frontend fixed; the validation therefore isolates the engine contribution to the pass-through, and it does not constitute a vendor-independent cascade, since the recognition stage is still the provider's audio-LLM. Table 6 reports the resulting two-by-two matrix of backends and

references. The diagonal cells, in which backend and reference share an engine, exceed the off-diagonal cells by 23 to 36 BLEU on clean speech (69.1 and 60.2 against 32.9 and 36.8), which is the same-family affinity of Section 4.6 measured on real speech; every backend looks strongest against its own reference, and the two off-diagonal cells differ by four BLEU points in favour of the GLM backend, so neither engine dominates decisively when scored away from its own family. The GLM backend's flatter profile under noise is also visible in its absolute scores.

Three observations follow. First, the degradation behaviour is governed by the recognition frontend, not the engine: both backends lose a nearly identical fraction of their clean quality as babble deepens (33.7 against 33.9 percent at 0 dB, 89.4 against 87.3 percent at -5 dB under their own references, with the cross-reference cells within a few points of these), so swapping the translation engine moves absolute scores by tens of BLEU points but barely moves the relative-loss trajectory. Second, refitting the pass-through relation on the GLM backend's three non-clean operating points gives slopes of 0.34 (Mandarin) and 0.55 (English), against 0.38 and 0.72 for the provider engine on the same conditions – the English coefficient about a quarter shallower – and 0.43 and 0.72 for the nine-condition real-speech fit of Section 4.4; the relation is therefore reproduced on a disjoint engine at the same order of magnitude (a three-point fit, whose per-point correlation carries no inferential weight at $n = 3$), with the GLM backend's shallower slopes consistent with its flatter absolute-quality profile under noise. Third, the caveat cuts both ways: the second backend is measured on three non-clean operating points of one frontend only, so it establishes that the pass-through is not a single-engine artefact, not that the coefficient is engine-independent. For deployment planning the practical reading is that the recognition stage, not the choice among competent translation engines, is where acoustic degradation is won or lost.

4.9. The streaming translation frontend as a recogniser

The streaming translation service exposes its internal recognition output, which allows its frontend, a specialised streaming recogniser, to be scored as a seventh system on the same 144-sentence sweep subset as the directly purchasable systems, which is the comparison reported here. Its clean-condition error rate of 1.0 percent sits between the LLM-ASR systems at 0.4 and the general-purpose streaming specialist at 2.2 percent, a difference that is not individually significant, and under noise it tracks the LLM-ASR class rather than its own product class: at 0 dB it errs at 25.4 percent against 24.1 and 22.6 percent for the two LLM-ASR systems, with overlapping confidence intervals and no significant paired difference (Wilcoxon $p = 0.19$), while the general-purpose streaming specialist errs at 59.2 percent on the same sentences. At 2.5 dB it errs at 4.0 percent against 4.5 and 2.6 for the LLM systems. Two coverage caveats apply: the frontend is only observable where the translation session completed, so its sweep coverage is partial in some conditions (52 to 134 of 144), and its zero-error 5 dB figure in the supplementary material rests on 52 completed sessions and should not be read as outperforming its own clean score. The observation remains noteworthy for practitioners: a recogniser competitive with the best purchasable ASR of this provider under severe noise is not available as a recognition service but is embedded inside a translation product, which suggests that providers optimise internal streaming frontends for exactly the acoustic conditions where end-to-end translation must survive.

4.10. Sentence-level variability and latency

Fig. 7 shows the sentence-level error distributions for the strongest LLM recogniser and the specialised streaming recogniser on five representative conditions. The distributions are strongly right-skewed: under moderate degradation most sentences remain nearly correct while a tail of sentences fails completely, which is why the confidence intervals in Table 3 are wide relative to their means. Under 0 dB babble the median sentence is still recognised reasonably well by the LLM systems while the upper quartile fails outright, a heterogeneity

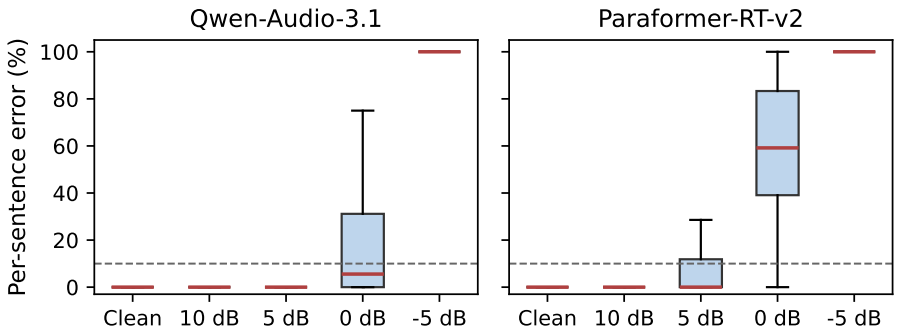


Figure 7: Sentence-level error distributions by condition for the strongest LLM recogniser and the specialised streaming recogniser.

that aggregate error rates conceal and that matters for applications with per-utterance quality requirements. The skew also explains a statistical property visible throughout Table 3: the confidence intervals of the LLM systems under moderate degradation are wide relative to their means because a small number of complete failures, including the refusal cases, coexist with a large mass of near-perfect transcripts. For a system operator this shape matters more than the mean, since the tail sentences are the ones that generate complaints and downstream errors, and the fraction of sentences above a given error threshold is a more actionable deployment metric than the average. On the 0 dB condition, the fraction of English sentences with error rates above 50 percent is 83 percent for the specialised streaming recogniser against 36 percent for Qwen3-ASR, while in Chinese the same threshold is exceeded by 33 percent of sentences for the specialised system against 3 percent for Qwen-Audio, so the family gap in tail behaviour is considerably larger than the gap in means.

Table 7 reports whole-utterance turnaround latencies. The text translation stage, benchmarked separately with 120 sequential timed calls, completes in 0.35 seconds per sentence on average (median 0.26, 95th percentile 0.46), so the fastest cascade that this provider can assemble, Qwen3-ASR feeding the translation engine, completes in about 0.7 seconds per sentence on average, the Qwen-Audio cascade in about 1.2 seconds, and the omni end-to-end path aver-

ages 2.8 seconds with a 95th percentile above 7 seconds, reflecting the substantially longer generation path of an omni-modal foundation model that produces its translation in a single response. The streaming translation service completes in about half a second per utterance. These figures are whole-utterance turnaround, not first-partial latency: the streaming interfaces would show substantially lower time-to-first-text than the whole-utterance numbers suggest, so latency comparisons across interface classes should be treated as indicative. Latency therefore reinforces rather than reverses the accuracy picture: the paths that degrade least under noise, the well-frontended cascades and the streaming translation service, are also among the fastest, while the slowest path, the omni end-to-end model, is the most fragile of the end-to-end paths and degrades more than the LLM-frontend cascades under severe noise in this measurement.

The repetition runs quantify how much of the measured variation is run-to-run noise rather than signal. Re-running the two LLM-ASR systems twice on the 0 dB babble and tempo conditions yields condition-level error rates that vary by at most 0.9 percentage points across three runs (0 dB: 24.1, 24.9, and 24.4 for Qwen-Audio; 22.6, 23.4, and 23.2 for Qwen3-ASR), while individual transcripts agree in 79 to 99 percent of sentences between runs. Condition-level conclusions in this paper are therefore stable against decoding nondeterminism at default settings, but sentence-level deployments should expect one in five transcripts to differ between identical calls at the worst condition. The omni path's small packet-loss and tempo anomalies of Section 4.5, two to three BLEU points, exceed any error-rate variation we can measure on the systems we could re-run, but the omni system's quota was exhausted before it could be re-tested, so its anomaly remains unexplained rather than confirmed as genuine.

5. Discussion

5.1. Guidance for system integrators

The measurements translate directly into selection guidance for practitioners who compose commercial speech services. Above a signal-to-noise ratio of

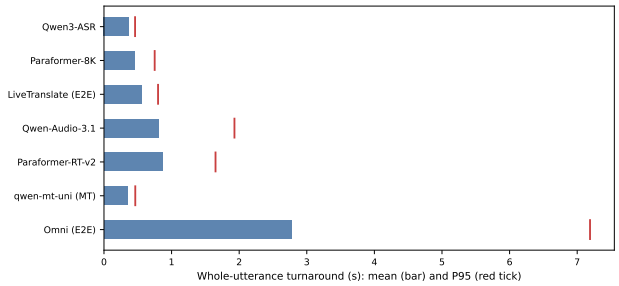


Figure 8: API round-trip latency by system, mean and 95th percentile.

about 10 dB, every recognition system measured here operates within one point of its clean error rate on clearly articulated speech, and the choice among them can be driven by latency, cost, and interface ergonomics rather than accuracy. The latency data in Table 7 then favour the audio-input LLM systems and the specialised streaming recognisers, which all answer in well under a second, over the omni-modal path. Between 0 and 5 dB, the product classes separate, and the knee thresholds of Table 4 express the difference most usefully: the LLM-ASR systems cross ten percent error 2.2 to 2.5 dB later in Mandarin and 4.4 to 4.8 dB later in English than the specialised streaming system, so in intermittently noisy environments such as public spaces, vehicles, or shared offices, the foundation-model recognisers hold the better end of the accuracy-latency trade-off, with the real-speech caveat that the advantage is confined to the near-knee regime and reverses for one LLM system in the hallucination regime (Section 4.4). Below 0 dB, every system fails, the LLM systems by hallucination and the streaming systems by silence, and the appropriate response is architectural: capture improvement, multi-microphone processing, or explicit rejection logic. For telephone-band audio, the surprising result is that no specialist premium exists: the telephone specialist matches neither the generalists' accuracy in its own band nor their robustness elsewhere. Its reverberation fragility, 18.2 percent error against at most 3.2 percent for every other system, proves specific to the synthetic extreme-DRR response: under three measured room impulse responses at matched reverberation time it degrades only from 5.1 to 6.3 per-

cent error (Section 4.4). Its selection can be justified only by its 8 kHz interface requirements and its speed, not by accuracy.

For translation pipelines, the frontend choice dominates the measurements reported here, and the latency data combine with it into a compact decision rule. A cascade assembled from the fastest recognition tier and the text translation engine completes in about 0.7 seconds per sentence on average, roughly four times as fast as the omni end-to-end path, while the streaming translation service completes in about half a second and is therefore the natural choice when latency dominates; when noise robustness dominates, the same streaming service and the LLM-frontend cascades lead the field, so the two objectives point to the same shortlist. Swapping the recognition stage of a cascade from the specialised streaming system to the LLM recogniser is worth 28 BLEU points at 0 dB (44.1 against 15.8) on an identical translation backend, a far larger effect than any measurement noise in this study, and the second-backend validation of Section 4.8 shows that swapping the translation engine instead, while holding the frontend fixed, barely moves the relative-loss trajectory, so within the measured class the frontend is the dominant term. The pass-through regression of Section 4.7, at roughly half a BLEU point per percentage point of recognition error with an intercept that is statistically indistinguishable from zero and numerically within a point of it, allows an integrator to rank candidate frontends with recognition-only measurements, which are cheaper to run and easier to interpret than full-pipeline evaluations, and to forecast translation quality degradation from a measured or contracted recognition error rate, with the caveats that the main-grid coefficient is calibrated on one translation engine (the second-backend three-point fit on a disjoint engine gives 0.34 to 0.55 on the shared conditions), that its honest uncertainty is the condition-level interval rather than the sentence-level one, and that the linear form saturates in the whole-utterance failure regime. The end-to-end versus cascade decision should rest on measured degradation behaviour rather than architectural intuition: the omni-modal end-to-end path evaluated here is simultaneously the slowest and, under severe noise, the most fragile of the end-to-end paths and

more fragile in point estimate than the LLM-frontend cascades, while a dedicated streaming translation service achieves both the smallest relative loss at 0 dB and sub-second latency.

5.2. Why foundation-model recognisers degrade more slowly

The robustness advantage of the LLM-ASR systems echoes the finding of Radford et al. [9] that recognition models trained on very large and weakly curated corpora acquire substantial noise tolerance without explicit multi-condition training. Two of our observations sharpen this picture for API consumers. First, the advantage is strongly condition-dependent: it is large under energetic babble noise and negligible under codec, packet-loss, clipping, and tempo distortions, which barely affect any system. This pattern is consistent with robustness arising from exposure to diverse recording conditions during pre-training rather than from any explicit enhancement front-end. Second, the advantage is language-asymmetric in these measurements, with Chinese recognition remaining usable at 0 dB while English recognition of the same systems fails at 37 to 50 percent, and, as Section 4.3 notes, part of the contrast may reflect the character-level versus word-level scoring scales rather than raw robustness alone. The Whisper baseline of Section 4.4 shows that this tolerance asymmetry is an engine property rather than a general law – the open-weights model is dominated by Mandarin errors already at +5 dB real babble where its English output errs at 19.3 percent – whereas the asymmetry that does replicate across engines is one of failure structure, Mandarin insertion-dominated hallucination against English deletion-dominated loss. Providers disclose neither training mixture nor per-language capacity, so the asymmetries can only be documented, not explained, at the black-box level; the documentation itself is actionable, since it implies that multilingual deployments should be derated per language and per engine rather than per provider. It also implies a concrete procurement test: a noise probe around each system's knee, a few hundred sentences of the target language in the 0 to 5 dB window, places a candidate service on the robustness scale documented here, and repeating the probe per deployment language

guards against extrapolating a strong Mandarin result to a weak English result or the reverse. The direction decomposition of Section 4.5 extends the same caution to translation deployments, where the two directions of a bilingual service degraded by asymmetric amounts, from 1.3 to 1 ratio to 3.2 to 1, under identical noise.

5.3. Cascade versus end-to-end revisited

The classical comparison of Bentivogli et al. [12] found end-to-end and cascade systems competitive on clean speech with different error structures. Our measurements extend that comparison to degraded channels in the black-box setting and qualify it in two ways. The end-to-end advantage in absolute quality against the same-vendor primary reference is real but partly a family artefact, since the omni system that translates directly also shares a generation family with the strongest reference producer, and the advantage inverts under the oracle reference and shrinks under the family-disjoint GLM reference, as Section 4.6 shows. More importantly, robustness is not an architectural property but a property of the acoustic frontend wherever it lives. The end-to-end service with the specialised internal frontend is the most resilient path we measured, while the end-to-end omni path is the most fragile of the end-to-end paths and degrades more than the LLM-frontend cascades in point estimate, with a reference-dependent margin, and the same frontend ordering that explains cascade behaviour also explains end-to-end behaviour. The real-speech validation adds a measurement-side caution to this picture: its translation reference shares the cascade's engine, so the absolute BLEU gap it reports between the streaming end-to-end service and the cascade (27 to 28 points on clean speech) is a same-engine artefact of the kind that Section 4.6 documents, not a translation-quality ranking. The relative-loss comparison is itself reference-sensitive on real speech: the end-to-end service loses 41.6 percent of its clean score under 0 dB real babble against 33.7 percent for the cascade under the same-engine reference – a point-estimate difference whose paired bootstrap interval crosses zero – but under the family-disjoint GLM reference the two losses converge to 32.6 and

32.7 percent, so the real-speech data are consistent with, without establishing, the relative-fragility conclusion, and they license no absolute claim. For a consumer, the practical question is not which architecture to prefer but where the provider placed its best acoustic model, and the answer is discoverable only by measurement of the kind this benchmark performs.

5.4. Methodological implications for black-box evaluation

Three methodological observations generalise beyond the specific services tested. The three-reference protocol of Section 4.6, in which one reference is family-disjoint from every system under test, demonstrates that machine-generated evaluation references can reorder systems in a way that tracks reference-generator affinity rather than translation quality, and that degradation-relative conclusions are the stable currency of black-box comparison. The synthesis-based corpus design gives black-box studies an exact transcript by construction, which matters because at sub-percent clean error rates, transcription uncertainty in recorded corpora would be comparable to the effects being measured. Finally, the empty-output behaviour documented in Section 4.1, where three of the five systems return empty transcripts on a systematic subset of severely degraded utterances, through two distinct mechanisms, argues that black-box benchmarks should report response completeness alongside error rates, since a missing transcript and a wrong transcript have different downstream failure modes, and a silently empty one differs operationally from a protocol-level refusal.

5.5. A worked example of the pass-through rule

The pass-through relation is most useful when stated as an explicit calculation, including its intercept and its validity bounds. For the Qwen-Audio cascade the fitted relation is a loss of $0.44\,e + 0.6$ BLEU points at recognition error e percent, and for the specialist cascade $0.54\,e - 1.0$, where the intercepts are point estimates that are statistically indistinguishable from zero and are carried for arithmetic consistency with the fitted line. Suppose an integrator

operates that cascade in an environment that occasionally degrades to 10 dB, at which the recogniser errs at 0.8 percent on the sweep subset; the predicted translation loss is 0.9 BLEU points, which is below the clean-condition agreement between the two reference systems and therefore negligible, and consistent with the measured 1.0 point loss at that condition. If the same deployment used the specialised streaming recogniser, whose 10 dB error rate is 2.4 percent on the same sentences, the specialist-calibrated regression predicts $0.54 \times 2.4 - 1.0$, about 0.3 BLEU points, again matching the measured figure within a point. In the 0 dB regime the rule predicts a loss of $0.44 \times 24.1 + 0.6$, about 11 points, against a measured 9.8, and for the specialist frontend $0.54 \times 59.2 - 1.0$, about 31 points, against a measured 35.7, so the rule is accurate to within a few points in the collapse regime but, as Section 4.7 quantifies, it is an empirical interpolation over the measured conditions rather than a law: its condition-level uncertainty spans 0.23 to 0.45 for the Qwen-Audio cascade, its main-grid coefficient is calibrated on one translation engine (the second-backend three-point fit reproduces the relation at 0.34 to 0.55 on the shared real-speech conditions), and it must not be extrapolated to recognition errors beyond the measured range, where BLEU loss saturates while recognition error can grow without bound. The same arithmetic applied to the measured error rates of a candidate system at the actual operating signal-to-noise distribution of a deployment yields an expected translation quality loss without running the translation stage at all.

5.6. Implications for evaluation practice and future work

Beyond the specific findings, the benchmark design carries lessons for future evaluations of commercial speech intelligence. A benchmark intended to track black-box services over time should log every raw response rather than only aggregate scores, because behavioural changes such as the refusal pattern documented here appear as distributional shifts before they appear in means. It should treat reference construction as a first-class experimental variable, since Section 4.6 shows that the reference generator can reorder systems. It should separate degradation gradients, which generalise across severity, from point es-

timates, which do not, and it should report response completeness alongside accuracy. The present study points to four directions for follow-up work: extension of the cross-vendor components beyond their current footholds – a full-strength non-vendor recogniser swept across the complete babble ladder and a second backend re-translating all nine real-speech conditions – replication across further providers to separate family effects from provider effects, extension to spontaneous and accented speech, and human-reference scoring of the translation stage to anchor the machine-reference comparisons. As foundation model services begin to expose streaming speech interfaces with embedded recognition, the measurement approach demonstrated here, treating each service as an opaque function from degraded audio to text and scoring the complete pipeline end to end, is, at present, the most direct methodology available for the comparative robustness questions that practitioners actually face.

6. Limitations

The benchmark uses synthetic speech from four neural voices, each reading each sentence once, so voice-level and take-level variability is not sampled and the corpus is read-style by construction. Synthetic speech lacks the disfluencies, accents, emotional variation, and channel noise of spontaneous human recordings, and although the exact-transcript property is decisive for measurement at sub-percent error floors, clean error rates measured on synthetic speech should not be read as population estimates for human speech; the real-speech validation of Section 4.4 addresses this concern for read-style recordings and reproduces the degradation structure, but it covers 150 utterances from two read-speech corpora, not spontaneous or accented speech, so the clean-condition levels remain indicative rather than population estimates. The most consequential design constraint is that the main grid – every recognition system, the synthesis, and two of the three references – comes from a single provider, so product class and vendor are fully confounded there: the class-level regularities reported here are consistent with published training-scale explanations [9], but cross-provider replica-

tion is required before they can be attributed to architecture rather than to one provider's engineering. The validation components loosen this confound only partway and only for structural conclusions: Whisper small is an open-weights model of the 2023 generation, executed locally on four of the nine real-speech conditions, so it serves as a failure-structure probe rather than a performance baseline and cannot localise its own knee within the tested range; the GLM backend swaps the translation engine on three non-clean operating points of a single frontend, enough to show that the pass-through is not a single-engine artefact but not enough to make the coefficient engine-independent; and every recognition frontend of every cascade remains the provider's, so no cascade measured here is vendor-independent. The reference situation improved but remains machine-anchored: the GLM reference is family-disjoint from every system under test, which removes the reference-affinity confound from relative-loss comparisons, yet all three references are still machine-generated, and neither human reference translations nor a learned reference-free metric (COMET class) was available within the zero-cost constraint, so absolute translation quality remains provisional; human references are the single addition that would most strengthen the absolute claims, although the real-speech validation sidesteps this concern for recognition, where its references are human transcriptions. The synthetic babble noise is smoother than real multi-talker cocktail-party interference, which includes semantic competition and stronger nonstationarity; the six-level real-babble ladder confirms that this gap makes synthetic babble the milder stressor, shifting the degradation knee upward by roughly 3–6 dB, without changing the system ordering or the collapse structure. Similarly, the reverberation condition uses a synthetic exponentially decaying response whose diffuse tail yields a direct-to-reverberant ratio of -27.6 dB, far below the $+7.3$ dB survey median for everyday spaces; the measured-impulse-response bridge condition shows real rooms at matched T_{60} to be markedly milder, so the synthetic reverberation results should be read as a stress case at the reverberant extreme. Severity levels for the non-noise degradations were fixed at deployed-but-aggressive values rather than swept, so only the noise family is characterised by a curve,

and the Opus and narrowband conditions proved to be partially redundant stressors at the chosen bitrate. Translation quality is measured with BLEU and chrF against machine-generated references, with the family-affinity bias that Section 4.6 quantifies; no learned reference-free metric (COMET class) or human judgement was available within the zero-cost constraint, and human-reference scoring is the single addition that would most strengthen the absolute claims. The corpus sentences were authored to be neutral and predictable, so the language-model prior of the LLM recognisers may partly mask acoustic degradation, and the numerals-domain gap may partly reflect inverse text normalisation rather than acoustics, as the normalisation sensitivity analysis suggests. Finally, commercial APIs are moving targets: the measurements document the October 2026 versions of the listed services, one system could not complete its full condition grid before its free-tier quota was exhausted, and the websocket transport of the streaming translation service dropped roughly ten percent of its sessions, which is itself a practically relevant datum for anyone planning a similar measurement on complimentary quotas.

7. Conclusions

This paper introduced CascadeBench, a black-box benchmark that measures how acoustic channel degradations propagate through commercial speech recognition and translation services, and reported a complete measurement over a controlled bilingual corpus of 480 sentences, fourteen degradation conditions from seven families, and an eight-level noise severity sweep from 20 to -10 dB, covering five recognition systems from three product classes, three cascade translation paths, and two end-to-end speech translation paths, all accessed through their public interfaces on free service quotas; the central design was validated on 150 human-recorded utterances under measured impulse responses and a six-level real-babble ladder, a measured-RIR bridge condition on the synthetic corpus, an open-weights locally executed recogniser, and a second-vendor translation engine used as both a family-disjoint reference and an alternative cascade

backend.

The measurements support five conclusions. First, the product classes of commercial recognisers differ in where they fail under noise, not merely how fast: LLM-based recognisers hold a sub-percent floor and cross ten percent error 2.2 to 2.5 dB later in Mandarin and 4.4 to 4.8 dB later in English than the specialised streaming system, then collapse more sharply, and below 0 dB they fail by insertion-heavy hallucination with character error rates above 100 percent; the real-speech validation reproduces this knee-and-collapse structure and the hallucination regime, with three calibrations: real rooms at matched reverberation time prove far milder than the synthetic low-DRR stress response (and a measured telephone-band specialist loses 3.4 rather than 18.2 percent under them), the LLM noise-tolerance advantage holds only in the near-knee, low-error regime – disappearing and for one LLM system reversing once error passes the ten-percent level – and is larger against real babble than synthetic babble, and the Mandarin-insertion, English-deletion failure structure replicates on an open-weights model outside the vendor, while language-specific tolerance levels do not. Second, robustness is condition- and language-specific rather than uniform: narrowbanding, coding, packet loss, clipping, and tempo changes at deployed severities are individually negligible for all systems, 0 dB babble nearly destroys English recognition for every system while leaving the best Mandarin recognition low, and no system, of any class, transcribes reliably below 0 dB. Third, the quality of a speech translation cascade under degradation is governed by its recognition frontend, and the relationship is an empirical linear pass-through of 0.44 to 0.54 BLEU points per percentage point of recognition error, with intercepts statistically indistinguishable from zero, stable under leave-one-condition-out refits for two of the three cascades, and honest condition-level uncertainty that is substantially wider than the sentence-level intervals, across up to fifteen conditions and three frontends; on real speech the same relation holds with slopes of 0.43 (Mandarin) and 0.72 (English) over nine conditions, and a three-point re-fit on a second vendor's engine reproduces the relation at the same order of magnitude (slopes 0.34 and 0.55 on the shared conditions),

an engine swap that moves absolute scores by tens of BLEU points but leaves the relative-loss trajectory nearly unchanged – a fit that establishes the relation is not a single-engine artefact, not that its coefficient is engine-independent. Fourth, end-to-end speech translation does not inherit robustness automatically: the most resilient translation path measured is a dedicated streaming end-to-end service, while the omni-modal end-to-end path is the slowest and the most fragile of the end-to-end paths, degrading more than the LLM-frontend cascades at 0 dB in point estimate under every reference, with a margin that shrinks to under one point against the Qwen3-ASR cascade under the oracle reference and whose paired interval crosses zero there; on real speech the streaming end-to-end path likewise loses a larger fraction of its clean quality under 0 dB real babble than a well-frontended cascade in point estimate under the same-engine reference, with a paired interval that crosses zero and losses that converge to equality under the family-disjoint reference, while the absolute BLEU ordering of the two paths remains reference-dependent. Fifth, black-box evaluation carries specific methodological obligations that this benchmark operationalises, including three-reference scoring with one family-disjoint reference to control machine-reference affinity – under which the relative-loss ranking is stable except for a sub-point inversion between the omni path and the Qwen3-ASR cascade under the same-engine reference – exact-transcript corpora to enable sub-percent measurement, the reporting of response completeness alongside error rates, and the anchoring of service-level findings in measured signal properties, which here exposed a sharp dissociation between intelligibility metrics and recognition error across degradation families.

The benchmark recipe, including the corpus, the seeded degradation pipeline, the degraded audio tree, all model responses, and the analysis code, is released so that the measurement can be re-executed and verified as providers update their services. The released artefacts are versioned alongside the service identifiers and access dates of every system, so that future re-runs can attribute score movements either to provider-side model updates or to changes in the benchmark itself, a distinction that periodic evaluation of moving targets requires.

Together with the zero-cost execution protocol described in Section 3, the release is intended to make robustness re-measurement a routine maintenance activity for speech deployments rather than a one-off research exercise. As speech intelligence continues to migrate into provider-operated black-box systems, controlled measurement of this kind, repeated over time and across providers, offers the community a practical substitute for the training-set access that classical robustness research assumed, and offers integrators the quantitative basis that architectural intuition alone cannot supply.

Appendix A. Corpus composition

Table A.8 summarises the corpus structure and lists one sample sentence per stratum. The four domains are constructed to differ in register, vocabulary, and numerical density. The broadcast-news domain uses formal written register with institutional vocabulary, the conversation domain uses colloquial everyday phrasing, the technical domain uses scientific and engineering terminology, and the numerals-and-entities domain concentrates dates, times, quantities, prices, addresses, codes, and named entities, with all numbers written out in words as described in Section 3. Length strata are defined per language as short, medium, and long as given in Section 3, and every domain-by-length cell contains twenty sentences per language, giving 240 sentences per language and 480 in total. The severity sweep subset of 144 sentences is sampled evenly across all cells, six sentences per cell per language.

Appendix B. Qualitative error analysis

Table B.9 collects representative model responses that illustrate the failure patterns behind the aggregate numbers in Section 4.

The refusal behaviour of Section 4.1 is shown in the first row: under 0 dB babble the Qwen-Audio recogniser returns an empty transcript for a substantial English sentence, which the benchmark scores as a complete failure. The second row shows the opposite regime: the same system, on a long English news

sentence at the same severity, produces a verbatim transcript, which underlines how bimodal the sentence-level distributions of Fig. 7 are. Row three shows the dominant error mode of the specialised streaming recogniser under severe noise, where most of the sentence is lost and only a short fragment survives. Row four shows a typical near-miss of the LLM recogniser in Chinese, where a single content word, density, is replaced by another word of similar phonetic shape, speed, without any structural damage. Row five shows the collapse of the telephone-band specialist under reverberation, which occasionally produces short disconnected fragments rather than transcripts. Row six shows an error that the recognition stage never sees but the translation consumer experiences: under 0 dB noise the omni end-to-end path substitutes a plausible but wrong entity, rendering the sports centre of the source as a grand theatre, a hallucination that a string-matching metric penalises only lightly but that is catastrophic for factual fidelity. Row seven shows the narrowband immunity of the LLM recogniser, which returns a verbatim transcript of telephone-band audio. Together the examples support the reading of Section 5: the systems differ less in whether they fail than in how, and the failure modes, from verbatim robustness to silent entity substitution to outright refusal, have distinct downstream consequences.

Appendix C. Cross-system text normalisation

Comparing transcripts across commercial recognisers requires a normalisation protocol that neutralises provider-specific inverse text normalisation. The protocol applies the following rules identically to references and hypotheses, in order. Unicode compatibility folding and case folding are applied first, and punctuation is removed. Numeric expressions are then canonicalised to digit form on both sides. For Chinese, the rules handle cardinal numbers with the standard multipliers, digit-by-digit readings such as room and code numbers where each digit is read separately, decimal fractions written with the point particle, time-of-day expressions in which the hour and minute parts are converted separately, percentage markers, which are deleted on both sides, and

scale multipliers, where a digit followed by wan, yi, or their combinations is expanded to the full multiplied-out form, so that twenty-six million written as 2600 万 in one system's output and as 26000000 in another's both canonicalise identically. For English, the rules handle cardinal numbers including compound hyphenated forms and the and connector, time and digit-by-digit readings such as twelve zero eight and nine forty, which canonicalise to 1208 and 940, decimal fractions expressed with point, scale multipliers from thousand to trillion, and ordinal expressions of both word and digit form, which are masked to a common placeholder since their spoken-to-written realisations differ across systems in ways that carry no acoustic information. The rule set was developed and frozen by inspecting all high-error sentences of the clean condition until no residual mismatches attributable to normalisation remained, a procedure that could in principle favour whichever system's conventions dominate the clean outputs, so a sensitivity analysis was run: re-scoring the main grid with case and punctuation folding only, dropping numeral canonicalisation entirely, raises clean-condition error rates from 0.5, 2.3, and 0.3 percent to 6.1, 6.2, and 0.8 percent for Qwen-Audio, Paraformer, and Qwen3-ASR respectively, while the 0 dB scores move by less than 1.5 points and the system ordering is unchanged in every condition. The normalisation therefore compresses but does not create the reported system differences, and the residual protocol noise on clean speech is bounded above by the strongest system's per-language clean rate of 0.14 percent in Mandarin (0.29 percent pooled over both languages), with the caveat that the two systems most affected by numeral conventions differ most under the raw protocol. The full normalisation code is released with the benchmark.

Appendix D. Six-level real-babble recognition ladder

Table D.10 reports the point-estimate recognition error of every system at each of the six real-babble signal-to-noise ratios of the real-speech validation, together with the interpolated ten-percent crossing level, so that the level- and language-dependent ordering discussed in Section 4.4 – including the reversal

of the Qwen3-ASR system against the two streaming specialists in Mandarin – can be read directly from a table in the main paper. Sentence-level bootstrap confidence intervals for the same cells, the clean, measured-reverberation, and transport-noise conditions, and the Whisper small decomposition are given in Supplementary Tables S5 and S7. The telephone-band specialist was evaluated on the Mandarin half only, and Whisper small on four of the six levels, as described in Section 3.4.

Data and code availability

The complete benchmark is publicly released at https://github.com/43aquarius/CascadeBench-Paper-AppliedAcoustics: the sentence corpus, the seeded perturbation implementations, the full degraded audio tree, all per-sentence model transcripts and translations (43,046 logged API responses), the analysis scripts, and the LaTeX source of this manuscript. Every number and figure in the paper is regenerated from the released logs by the released analysis scripts. Extended numeric tables (complete pairwise significance results, severity-sweep numerics, refusal tabulations, and the complete real-speech, Whisper, measured-RIR, and second-backend results) are provided as supplementary material. Because the journal operates single-anonymised review, in which author identity is visible to reviewers, the public link is given directly; an anonymised snapshot is additionally provided with the submission.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large language model assistant in the development of the analysis scripts and in drafting revisions of the manuscript text, in order to accelerate the benchmark construction and to improve the readability of the presentation. The experimental measurements, the analysis decisions, and the verification of all reported numbers against the

released logs were performed and checked by the authors, who take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 2: API call accounting in two counting conventions: successful calls (every logged successful response, re-runs included) and unique evaluations (distinct system–condition–sentence triples). Coverage notes: the 8 kHz specialist is evaluated on the Chinese subset; the omni system is restricted by quota exhaustion as described in the text; the measured-RIR bridge condition applies the three survey impulse responses to the synthetic corpus, so it is a synthetic-speech condition and is reported separately from the real-speech validation; Whisper runs are local and carry no API cost.

Phase	System	Calls	Unique	Coverage
Synthesis	Sambert (4 test + 6 babble voices)	520	520	480 + 40 babble
Recognition	Paraformer-RT-v2	5,174	4,752	full grid + sweep
Recognition	Paraformer-RT-8K	2,376	2,376	zh subset+sweep
Recognition	Qwen-Audio-3.1	5,154	4,752	grid+sweep (24 ref.)
Recognition	Qwen3-ASR	4,752	4,752	full grid + sweep
Recognition	Qwen3.8-Omni	1,271	1,271	main grid, sweep subset
Translation	oracle + primary reference	960	960	480 + 480
Translation	cascades (2 frontends)	3,456	3,456	12 conds × 144
Translation	E2E omni	1,004	1,004	83–84/144, 12 conds
Translation	E2E LiveTranslate	1,571	1,544	89.5% of 12 × 144
Original phase subtotal		26,238	25,363	
Extension	4 ASR × 3 new SNRs	1,512	1,512	2.5/−5/−10 dB × 144
Extension	cascades on new SNRs	858	858	2 frontends, 3 conds
Extension	Qwen3-ASR cascade (12 conds)	1,728	1,728	third cascade system
Extension	E2E LiveTranslate on new SNRs	251	251	251 of 432 targets
Extension	repetition runs (2 systems)	1,112	1,112	8 runs × 144
Extension	MT latency benchmark	120	120	timed calls
Extension subtotal		5,581	5,581	
Real validation	4 ASR × 6 core conds	3,150	3,150	150 utts, 8 kHz zh
Real validation	cascade (Qwen-Audio → MT)	900	900	150 utts × 6 conds
Real validation	oracle MT (true transcripts)	150	150	reference generation
Real validation	E2E LiveTranslate	450	450	3 conds × 150 utts
Real validation	ladder to 6 babble SNRs	2,025	2,025	3 added SNR levels
Real validation subtotal		6,675	6,675	
Measured RIR	4 ASR (synthetic corpus)	1,680	1,680	480×3 + 240 zh
Measured RIR	cascades, 4 frontends	1,680	1,680	incl. 240 8 kHz
Measured-RIR bridge subtotal		3,360	3,360	
Second vendor	GLM references	630	630	480 synth + 150 real
Second vendor	GLM backend re-translation	562	562	4 conds, Qwen-Audio
Second-vendor subtotal		1,192	1,192	
Total successful responses		43,046	42,171	
Failed attempts (3,333 LT drops + 541 API)		3,874	—	
Local Whisper runs (no API access)		—	—	600 transcriptions

Table 3: Mean recognition error rates in percent on the main degradation grid, both languages pooled, with bootstrap 95 percent confidence intervals in parentheses. The 8 kHz specialist is evaluated on the Chinese subset only.

System	Clean	Babble 10	Babble 0	Reverb	Narrow	Codec	Loss	Clip	Tempo
Paraformer-v2	2.3	2.8	58.5	3.2	2.1	2.4	3.2	2.5	2.3
	(2,3)	(2,4)	(54,63)	(2,4)	(2,3)	(2,3)	(2,4)	(2,3)	(2,3)
Paraformer-8K	1.9	4.7	73.2	18.2	1.9	1.8	3.3	1.9	2.3
	(1,3)	(3,6)	(69,78)	(14,22)	(1,3)	(1,3)	(2,5)	(1,3)	(2,3)
Qwen-Audio-3.1	0.5	0.9	28.2	0.7	0.6	0.6	0.6	0.9	0.6
	(0,1)	(1,1)	(25,32)	(0,1)	(0,1)	(0,1)	(0,1)	(1,1)	(0,1)
Qwen3-ASR	0.3	0.5	25.1	0.3	0.3	0.3	0.4	0.3	0.3
	(0,0)	(0,1)	(22,29)	(0,0)	(0,0)	(0,0)	(0,1)	(0,0)	(0,0)
Qwen3.8-Omni	0.4	0.6	31.8	1.1	0.1	0.3	0.2	1.0	0.3
	(0,1)	(0,1)	(28,36)	(1,2)	(0,0)	(0,0)	(0,0)	(1,2)	(0,0)

Table 4: Noise severity characteristics on the eight-level sweep: the signal-to-noise ratio at which the piecewise-linear interpolation of the measured curve crosses ten percent error, with sentence-level bootstrap 95 percent confidence intervals; the maximum slope of the fitted logistic as a knee-sharpness measure (the fitted knee of both LLM-ASR systems in Mandarin lies between 0 and −5 dB, where no observation level exists); the steepest observed segment slope between adjacent measured levels; and sub-range OLS slopes, listed for continuity with common practice and depending strongly on the fitted range.

	SNR@10% (dB)		Max slope		Steep. seg.		OLS 20–0	
System	zh	en	(pp/dB)		(pp/dB)		(pp/dB)	
Paraformer-RT-v2	2.48 (2.26,3.23)	6.79 (4.93,7.91)	20.8	16.1	13.0	15.2	−1.44	−2.97
Paraformer-RT-8K	8.82 (8.17,9.31)	n/a	10.4	n/a	9.9	n/a	−3.31	n/a
Qwen-Audio-3.1	−0.06 (−0.25, 0.68)	2.38 (2.02,3.38)	35.5	21.3	23.7	12.6	−0.26	−1.31
Qwen3-ASR	0.29 (−0.11, 0.92)	2.02 (1.73,2.31)	25.2	25.0	18.7	13.5	−0.35	−1.06

Table 5: Translation quality by condition on the sweep subset: macro-averaged BLEU against the primary reference (Qwen3.8-Max translations of clean transcripts) with sentence-level bootstrap 95% confidence intervals, the clean-to-0 dB relative loss under each of the three references (primary, oracle cascade output, GLM second-vendor), and the refusal-inclusive chrF at 0 dB. Cascade scores at 0 dB are conditional on a non-empty frontend output; absolute 0 dB scores against the oracle and GLM references are tabulated in the supplementary material. The omni path is scored on its quota-limited subset and is compared with the other paths on the common subset in the text.

System	Clean BLEU	0 dB BLEU (95% CI)	Rel. loss at 0 dB vs primary	vs oracle	vs GLM	chrF 0 dB
Oracle MT (clean text)	55.5	–	–	–	–	–
Cascade: Paraformer+MT	51.5	15.8 (12.3,19.1)	69%	80%	73%	26.9
Cascade: Qwen-Audio+MT	53.9	44.1 (40.1,48.3)	18%	24%	17%	53.7
Cascade: Qwen3-ASR+MT	53.2	43.0 (38.7,47.9)	19%	29%	23%	53.4
E2E: Omni-Flash	68.1	45.5 (38.2,50.5)	33%	29%	30%	51.6
E2E: LiveTranslate	59.8	50.4 (41.8,51.4)	16%	21%	17%	56.2

n: cascades 144 clean, 135–144 non-empty at 0 dB; omni 84; LiveTranslate 144/132; CIs from 300 sentence-bootstrap resamples; chrF refusal-inclusive.

Table 6: Second-backend validation on real speech: corpus BLEU of the Qwen-Audio cascade under both translation backends, scored against both validation references (oracle: the provider engine on human transcripts; GLM: the second vendor on the same transcripts). Diagonal cells share an engine between backend and reference. The final column gives the within-backend clean-to-−5 dB relative loss under the reference shown.

Backend	Reference	BLEU by condition Clean	+5	0	−5	Rel. loss to −5
qwen-mt-uni	oracle (same engine)	69.1	66.2	45.8	7.3	89%
qwen-mt-uni	GLM (cross)	32.9	31.2	22.1	4.9	85%
GLM	oracle (cross)	36.8	36.0	26.3	6.7	82%
GLM	GLM (same engine)	60.2	55.9	39.9	7.6	87%

n: 150 per cell except GLM backend 148 at 0 dB and 114 at −5 dB.

Table 7: Whole-utterance API turnaround latency in seconds for the recognition systems, the text translation stage (120 sequential timed calls), and the end-to-end paths. Streaming interfaces are measured at maximum feed rate; see text for the first-partial caveat.

System	Mean	P50	P95
Qwen3-ASR-Flash	0.37	0.35	0.46
Paraformer-RT-8K-v2	0.45	0.43	0.75
LiveTranslate (E2E, streaming)	0.55	0.52	0.80
Qwen-Audio-3.1-Flash	0.80	0.72	1.93
Paraformer-RT-v2	0.87	0.85	1.65
Text translation (qwen-mt-uni, n=120)	0.35	0.26	0.46
Qwen3.8-Omni-Flash (E2E)	2.77	2.20	7.19

Table A.8: Corpus structure with one sample sentence per domain.

Domain	Sample sentence
News (zh)	国家统计局今天发布了上月经济运行数据。
News (en)	The central bank raised interest rates by twenty five basis points to curb inflation.
Conversation (zh)	我打算下班之后去健身房锻炼一个小时。
Conversation (en)	I was thinking we could drive to the beach this weekend if the weather holds.
Technical (zh)	混响是声波在房间内多次反射形成的，会显著降低远场语音识别的准确率。
Technical (en)	Reverberation arises when sound waves reflect repeatedly off room surfaces.
Numerals (zh)	从北京到上海的高铁大约需要四个半小时。
Numerals (en)	The high speed train from Beijing to Shanghai takes about four and a half hours.

Table B.9: Representative system responses under degradation.

Condition	System	Reference and response
0 dB, en	Qwen-Audio-3.1	Reference: Police arrested two suspects on Tuesday. Response: empty transcript.
0 dB, en	Qwen-Audio-3.1	Reference: The transport minister told reporters on Friday that the government would invest heavily in electric vehicle charging infrastructure. Response: verbatim transcript.
0 dB, en	Paraformer-RT-v2	Reference: The battery lasts twenty hours. Response: About.
0 dB, zh	Qwen-Audio-3.1	Reference: 这款处理器的晶体管密度提升了近一倍。Response: 这款处理器的晶体管速度提升了近一倍。
Reverb, zh	Paraformer-RT-8K	Reference: 记者从发布会上了解到，本次合作项目将覆盖全国二十多个主要城市。Response: 七月同。非常了解到。
0 dB, E2E	Omni-Flash	Source: 新建成的体育中心可容纳六万名观众。Translation: The newly built Grand Theatre can accommodate 60,000 audience members.
Narrow, en	Qwen-Audio-3.1	Reference: This coffee shop is cozy. Response: verbatim transcript.
−10 dB, en	Qwen-Audio-3.1	Reference: The battery lasts twenty hours. Response: fluent unrelated Chinese text of several sentences (hallucination).

Table D.10: Recognition error rate in percent (CER for Mandarin, WER for English) on the six-level real-babble ladder of the real-speech validation ($n = 75$ sentences per language per level). Cross-10% is the interpolated babble SNR at which error reaches ten percent; $> X$ means error already exceeds ten percent at the mildest level at or above X dB, so the crossing lies outside the measured range. Dashes mark levels not evaluated for that system. Confidence intervals for these cells are given in Supplementary Table S5.

	Real-babble SNR (dB)						Cross-10%
System	+10	+5	+2.5	0	−2.5	−5	(dB)
Mandarin							
Qwen-Audio-3.1	2.0	4.7	10.7	46.9	107.3	150.0	2.8
Qwen3-ASR	2.4	15.0	62.3	117.5	173.0	207.6	7.0
Paraformer-v2	12.1	40.1	64.8	102.6	131.3	164.4	$> +10$
Paraformer-8K	27.9	48.6	60.2	78.7	86.8	93.1	$> +10$
Whisper small	—	78.7	—	120.3	—	130.7	$> +5$
English							
Qwen-Audio-3.1	1.8	3.1	8.9	26.0	72.3	86.6	2.3
Qwen3-ASR	2.8	5.6	13.7	24.2	43.5	64.1	3.6
Paraformer-v2	9.5	24.7	39.5	56.7	76.8	86.1	9.8
Whisper small	—	19.3	—	47.7	—	92.3	$> +5$